

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

HUMAN RESOURCE DEPARTMENT WEB-BASED JOB PORTAL SYSTEM
FOR NATIONAL AVIATION ACADEMY OF THE PHILIPPINES

An Undergraduate Capstone Project Presented to the Faculty of the
Institute of Computer Studies

NATIONAL AVIATION ACADEMY OF THE PHILIPPINES

Piccio Garden, Villamor, Pasay City

In Partial Fulfillment of the Requirements for the Degree of
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY WITH THE
SPECIALIZATION AVIATION INFORMATION TECHNOLOGY

ALARCIO, CHARLES JOHN S.

CACACHO, MARCUS ANGELO T.

DR. ALAN LINO SILVERIO J AGUSTIN, DEAM

SEPTEMBER 2026

CHAPTER 1

INTRODUCTION

Recruitment requires an organized process for publishing vacancies, receiving applications, checking qualifications, communicating application status, protecting applicant records, and preparing recruitment reports. Web-based recruitment, applicant-tracking systems, and digital recruitment platforms connect these activities into a single process. [5], [7]– [14] A Philippine institutional applicant-tracking system was developed to replace manual storage and reporting, provide real-time status monitoring, and support decision-making. [8] Studies of online hiring platforms also report more convenient access to vacancies, better process transparency, and a better applicant experience. [9], [10] [1–9]

The review of related literature supports seven areas required for a Human Resource Department web-based job portal: AI-based candidate screening and qualification scoring; applicant registration, digital application, and status monitoring; digital transformation of recruitment; job-vacancy management; recruitment analytics and report generation; software-quality evaluation; and user authentication, role-based access control, and privacy. AI can assist the initial comparison of applicant information with vacancy requirements, but it must be designed with fairness, transparency, rights, and human review in mind. [1]– [6]

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

HR portals must also protect applicant records through authenticated, role-based access and appropriate access-management controls. [21]– [26] [10–20]

The reviewed literature supports the separate functions of a recruitment portal, but it does not provide a portal designed for NAAP that integrates all six required recruitment features and evaluates the completed system through the nine adopted ISO/IEC 25010:2023 quality characteristics, candidate-screening time, and HR-validated automated qualification-scoring accuracy. [8], [15]– [20] This gap provides the basis for the present study, entitled ***“Human Resource Department Web-Based Job Portal System for National Aviation Academy of the Philippines.”*** The AI-assisted score is only one input to the screening process; final hiring decisions remain with HR personnel, who review the complete application, verify the stated vacancy requirements, and apply institutional recruitment procedures. [3, 14, 21–26]

Statement of the Problem

The National Aviation Academy of the Philippines needs a recruitment process that manages vacancies, applications, qualification review, applicant status, access control, and recruitment information as one connected workflow. This study therefore aims to design, develop, implement, and evaluate a ***Human Resource Department Web-Based Job Portal System for National Aviation Academy of the Philippines***

Research Questions

This study seeks to answer the following questions:

1. What are the functional requirements of the proposed Human Resource Department Web-Based Job Portal System?
2. What are the non-functional requirements of the proposed system?
3. What features should be developed for the proposed system in terms of:
 - 3.1 job-vacancy management;
 - 3.2 applicant registration and digital application;
 - 3.3 AI-based candidate screening and qualification scoring;
 - 3.4 applicant tracking and status monitoring;
 - 3.5 user authentication and role-based access control; and
 - 3.6 recruitment analytics and report generation?
4. What is the level of software quality of the proposed system based on ISO/IEC 25010:2023 in terms of:
 - 4.1 functional suitability;
 - 4.2 compatibility;
 - 4.3 performance efficiency;
 - 4.4 interaction capability;
 - 4.5 reliability;
 - 4.6 security;
 - 4.7 maintainability;
 - 4.8 Flexibility; and
 - 4.9 safety
5. How does the proposed system perform in terms of:
 - 5.1 candidate-screening time; and
 - 5.2 automated qualification-scoring accuracy?

Project Context

Higher education institutions increasingly use web-based systems to organize administrative work and make information accessible to authorized users.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Recruitment is one of these activities because it requires institutions to publish vacancies, receive applications, review applicant qualifications, communicate results, and retain recruitment records.

This need is more specific in aviation education. Positions in an aviation academic institution may require defined educational qualifications, civil-service eligibility, professional experience, training, certifications, and technical credentials. Recruitment must therefore give HR personnel a clear way to state vacancy requirements and assess whether submitted applicant information meets them.

The project is situated in the Human Resource Department of the National Aviation Academy of the Philippines. NAAP recruitment requires HR personnel to manage vacancies, receive applicant requirements, review qualifications, update application status, protect applicant records, and prepare recruitment reports. Applicants need a clear way to view vacancies, submit documents, and monitor their application progress.

The current recruitment need is not met by one connected portal within the project scope. Vacancy information, applications, qualification checks, status updates, and reports must be handled as parts of the same recruitment process. Without a single system, recruitment work becomes harder to track, applicant status becomes less visible, and HR records become more difficult to organize.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

The literature reviewed confirms the value of digital recruitment, applicant tracking, AI-assisted screening, access control, reporting, and software-quality evaluation. However, it does not show one NAAP-specific portal that combines these functions and evaluates the completed system using the nine adopted ISO/IEC 25010:2023 characteristics, candidate-screening time, and automated qualification-scoring accuracy. The proposed Human Resource Department Web-Based Job Portal System is needed to address this institutional requirement.

Project Purpose and Description

This study develops and evaluates a recruitment portal for NAAP's Human Resource Department. It aims to provide an organized recruitment process that manages vacancy information, applicant submissions, qualification review, application status, access control, and recruitment reports in one system.

The proposed system is a web-based job portal with six recruitment features: job-vacancy management; applicant registration and digital application; AI-based candidate screening and qualification scoring; applicant tracking and status monitoring; user authentication and role-based access control; and recruitment analytics and report generation. The AI component compares submitted applicant information with vacancy criteria and produces a score to support HR review. It does not make a final hiring decision.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

The completed system will be evaluated through software-quality and operational-performance measures. Software quality will be assessed using functional suitability, compatibility, performance efficiency, interaction capability, reliability, security, maintainability, flexibility, and safety. Operational performance will be assessed through candidate-screening time and HR-validated automated qualification-scoring accuracy.

The system serves applicants, HR personnel, and authorized administrators. Applicants may view vacancies, create an account, submit applications and required documents, and monitor their own application status. HR personnel may manage applications, review AI-assisted qualification results, update application stages, and prepare recruitment reports. Authorized administrators may manage vacancies, reports, and user roles. The system is limited to recruitment and applicant management; it does not cover payroll, attendance, onboarding, or employee performance management.

The study will provide NAAP with a recruitment-focused system that organizes recruitment records, makes applicant status visible, controls access to applicant information, and provides reports for authorized HR use. It also contributes an evaluated example of a web-based recruitment portal for a specialized aviation academic institution.

Objectives of the Study

General Objective

To design, develop, implement, and evaluate a Human Resource Department Web-Based Job Portal System for the National Aviation Academy of the Philippines (NAAP) that automates recruitment processes, facilitates applicant screening and tracking, and supports efficient and secure management of recruitment information.

Specific Objectives

1. To determine the functional requirements of the proposed system.
2. To determine the non-functional requirements of the proposed system.
3. To design and develop the proposed system with the following recruitment and applicant-management features:
 1. job-vacancy management;
 2. applicant registration and digital application;
 3. AI-based candidate screening and qualification scoring;
 4. applicant tracking and status monitoring;
 5. user authentication and role-based access control; and
 6. recruitment analytics and report generation.
4. To evaluate the software quality of the proposed system based on ISO/IEC 25010:2023 in terms of functional suitability, compatibility, performance efficiency, interaction capability, reliability, security, maintainability, flexibility, and safety.
5. To evaluate the operational performance of the proposed system in terms of candidate-screening time and automated qualification-scoring accuracy.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presents a thematic review of literature relevant to developing a web-based, AI-assisted job portal system for the Human Resource Department of the National Aviation Academy of the Philippines (NAAP). It draws on literature and technical publications from 2022–2026. The themes are arranged alphabetically and support the proposed portal features, software-quality evaluation, candidate-screening time, and automated qualification-scoring accuracy.

AI-Based Candidate Screening and Qualification Scoring

AI-based candidate screening uses digital methods to compare applicant information with the qualifications required for a vacancy. Qualification scoring turns this comparison into a score or rank that helps HR personnel review applications. It is a support tool, not a replacement for HR judgment.

Mori *et al.* review 120 articles and frame AI recruitment around efficiency, justice, and rights [1].[1] Kassir *et al.* discuss efficiency, objectivity, and consistency as proposed benefits of machine-learning hiring tools, while emphasizing the need to address disparate impact [2].[2] Dadaboyev *et al.* identify

résumé parsing, chatbot screening, predictive job-fit analysis, and AI-assisted interviews as recruitment uses [3].[3] González-Morales *et al.* similarly identify pre-selection, résumé classification, interview analysis, skill assessment, and job-suitability prediction [4].[4] Putra *et al.* link applicant tracking systems, online assessment, video interviewing, and AI to digital recruitment work [5].[5] At the system level, Vineela *et al.* tested a hybrid NLP and ensemble approach to résumé screening and reported better performance for the ensemble than for its individual classifiers [6].[6]

The six sources agree that AI can reduce repetitive work in initial screening. They differ in their emphasis: reviews identify ethical and organizational conditions, whereas system studies test matching or classification models. The evidence does not justify treating an automated score as the final hiring decision. For NAAP, the module should score applicants against published vacancy requirements, show the basis of the score, and require HR review before an applicant moves to the next stage. Its operational evaluation should compare candidate-screening time with the existing procedure and compare each automated qualification result with HR personnel's documented application of the vacancy criteria. [1, 2, 6]

Applicant Registration, Digital Application, and Status Monitoring

Applicant registration allows a job seeker to create an account, provide profile and qualification information, upload required documents, and submit an

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

application online. Status monitoring makes the application's progress visible to both the applicant and authorized HR personnel.

Khan and Kawadkar report that applicants viewed e-recruitment as saving time and cost while expanding employment access [7].[7] Jacobe and Pascua developed an applicant tracking and management system with real-time monitoring, alerts, applicant summaries, and comparative reports in a Philippine institutional setting [8].[8] Adawiah and Putra find that e-recruitment affected perceived organizational credibility, selection efficiency, and process transparency from job applicants' perspective [9].[9] Sandhu *et al.* report that surveyed job seekers viewed online hiring platforms as more convenient, transparent, and effective, while reducing time spent searching [10].[10] Jain and Pandey describe digital platforms as supporting candidate sourcing, screening, and selection while raising questions about bias and candidate experience [11].[11] Chavan *et al.* describe ATS functions spanning application receipt, screening, testing, scheduling, and hiring-process management [12].[12]

The sources support one online record for each applicant and clear recruitment-stage updates. They differ between applicant surveys, institutional system development, and broader studies of digital recruitment. For NAAP, applicants should be able to register, complete a digital application, upload

required files, and see understandable updates such as *submitted, for review, qualified/not qualified, for interview, and closed*. [8, 10, 12]

Digital Transformation of Human Resource Recruitment

Digital transformation of human resource recruitment uses web-based systems to manage vacancy posting, application submission, screening, communication, tracking, and reporting. It brings related recruitment work into a connected digital process.

Putra *et al.* associate digital recruitment and selection tools with efficiency and recruitment quality [5].[5] Khan and Kawadkar document applicants' preference for e-recruitment processes [7].[7] Adawiah and Putra link e-recruitment implementation with applicant-facing transparency and selection efficiency [9].[9] Sandhu *et al.* provide applicant-side evidence that online platforms can improve access to vacancies and the job-search experience [10].[10] Jain and Pandey examine the role of digital platforms and emerging AI tools in recruitment sourcing, screening, and selection [11].[11] Inchulkar and Dubey describe job portals, social media, ATS, and AI tools as digital recruitment platforms, while also noting privacy, technical, and human-interaction concerns [14].[13]

The literature supports moving connected recruitment activities online, but it comes from different sectors and countries rather than a state aviation

academy. It also shows that digitalization does not remove the need for policy, privacy protection, and human decision-making. For NAAP, the web portal should consolidate recruitment records and workflow rather than rely on separate paper forms, email threads, and spreadsheets. [9, 13]

Job Vacancy Management

Job vacancy management is the process of creating, updating, publishing, monitoring, and closing job vacancies. A web-based portal gives HR personnel a controlled way to present position requirements and deadlines, receive applications, and retain a history of recruitment activity.

Jacobe and Pascua demonstrate the use of structured applicant records and reporting within an institutional tracking system [8].[8] Adawiah and Putra connect online recruitment with selection-process transparency [9].[9] Sandhu *et al.* find that web-based hiring platforms can make job opportunities more accessible and transparent to applicants [10].[10] Jain and Pandey include sourcing, screening, and selection among the functions supported by digital recruitment platforms [11].[11] Chavan *et al.* describe ATS as a system for electronically managing recruitment needs from application receipt through hiring [12].[12] R. and Rajeswari describe a web-based job portal in which recruiters post openings, manage applications, and shortlist candidates [13].[14]

These sources support connecting vacancy publication to application handling and recruitment records. They do not establish one standard vacancy form suitable for every institution. The NAAP portal should therefore let authorized HR personnel create and edit a vacancy with its position title, qualification criteria, deadline, publication status, and closing record. [12, 14]

Recruitment Analytics and Report Generation

Recruitment analytics organizes vacancy and applicant data into information that HR personnel can monitor. Report generation presents that information in summaries that support recruitment planning, workload monitoring, and record keeping.

Putra *et al.* discuss digital recruitment tools that connect recruitment and selection activities [5].[5] Jacobe and Pascua incorporate dashboards, applicant summaries, comparative reports, and institutional decision support in their system [8].[8] Adawiah and Putra identify transparency and selection efficiency as e-recruitment outcomes relevant to monitoring recruitment processes [9].[9] Sandhu *et al.* show that online hiring platforms affect the applicant experience and access to job opportunities [10].[10] Jain and Pandey describe digital platforms as supporting the recruitment process from sourcing through selection [11].[11] Inchulkar and Dubey connect digital recruitment platforms with time-to-hire, cost, communication, and hiring outcomes [14].[13]

The sources support using organized recruitment records to observe activity and produce reports, but the reports they emphasize vary by setting. For NAAP, recruitment reports should present vacancy status, application counts per vacancy, applicant-stage distribution, and processing time by stage. Each report should identify its date range and filters so that HR personnel can interpret it correctly. [8, 13]

Software Quality Evaluation for Web-Based HR Systems

Software-quality evaluation determines whether a web-based portal provides the required functions correctly and performs with appropriate efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety. It evaluates the system beyond the mere presence of its features.

ISO/IEC 25010:2023 defines a product-quality model with nine characteristics that can be specified, measured, and evaluated for ICT and software products [15]. [15] Rojas *et al.* map research use of the standard across public administration, finance, education, and healthcare while identifying implementation challenges [16]. [16] Jereb *et al.* apply ISO/IEC 25010:2023 to operational information systems [17]. [17] Kim examines use of ISO/IEC 25010 for assessing web-based ERP-system completeness [18]. [18] Prameswari and Setyawan combine ISO/IEC 25010 technical dimensions with website evaluation and report findings on functional suitability, usability, reliability, and security

[19].[19] Oktariani and Utami integrate ISO/IEC 25010:2023 with OWASP Top Ten in a web-application evaluation and identify function and security issues together [20].[20]

The sources agree that web systems should be evaluated across quality characteristics rather than only by feature completion. Their contexts differ, so their results cannot be transferred directly to NAAP. The proposed portal should be evaluated using the nine ISO/IEC 25010:2023 characteristics adopted by the study: functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety. [15, 19, 20]

User Authentication, Role-Based Access Control, and Privacy

User authentication verifies the identity of a person requesting access. Role-based access control assigns functions and data access based on that person's authorized role, while privacy safeguards applicant information from unauthorized access, disclosure, or misuse.

NIST's Cybersecurity Framework 2.0 provides an official framework for managing cybersecurity risk [21].[21] NIST SP 800-63B-4 specifies authentication and authenticator-management requirements at defined assurance levels [22].[22] CISA's administrator guidance treats identity and access management as a major control area [23].[23] CISA's ICAM reference architecture provides a government framework for identity, credential, and access management [24].[24] Sahyudi and

Susanto review RBAC implementation and identify privacy, compliance, and access-policy complexity as important concerns [25].[25] Purba and Hidayasari implement RBAC in a web-based information system and report that it restricts privileges according to authority while improving data-security control [26].[26]

The sources agree that access should be based on verified identity and authorized role. Government technical guidance provides broad security direction, while the RBAC studies concentrate on role and privilege implementation. For NAAP, applicants should see only their own applications; HR personnel should process recruitment records; and authorized HR administrators should manage vacancies, reports, and user roles. [21, 25, 26]

Synthesis

The reviewed literature supports a connected web-based recruitment portal that manages vacancies, digital applications, applicant status, AI-assisted screening, role-based access, and reports. It also shows that automated screening should use transparent vacancy criteria and HR oversight, while portal quality and security should be tested in the organization's actual working environment. [1, 2, 15, 21]

Sources differ in country, sector, technical scope, and method. The literature does not provide a single NAAP-designed portal that combines all six recruitment features and evaluates the completed system using ISO/IEC 25010:2023,

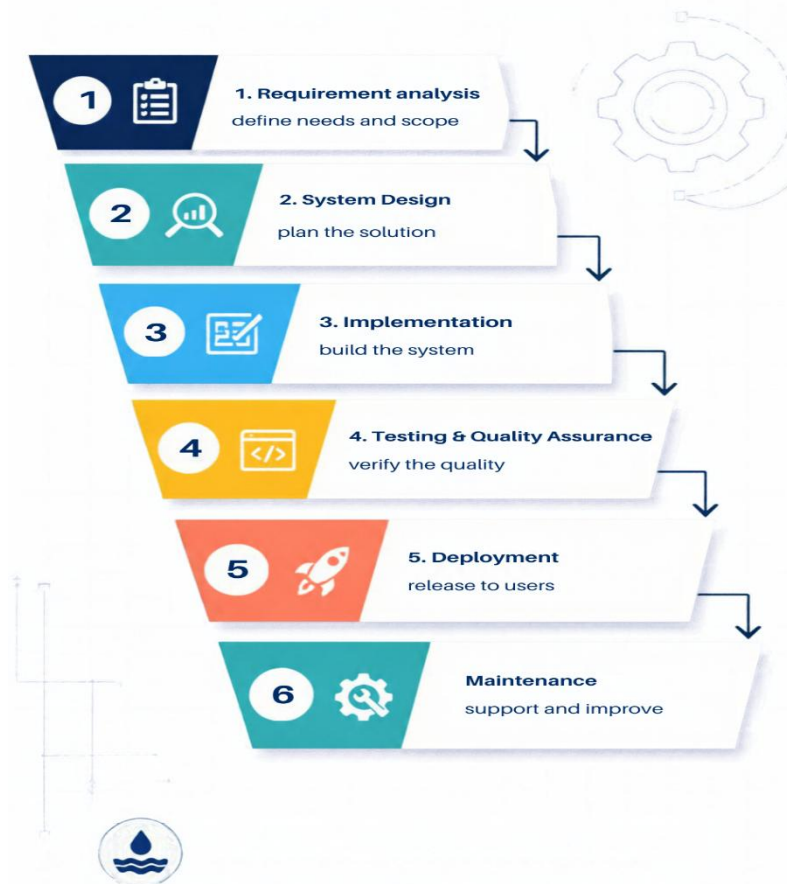
candidate-screening time, and HR-validated qualification-scoring accuracy. The present study addresses that institutional gap. [6, 8, 15]

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

Technical Background

The Human Resource Department Web-Based Job Portal System for National Aviation Academy of the Philippines is a browser-accessible recruitment system designed to organize vacancy management, digital applications, AI-assisted candidate screening and qualification scoring, applicant tracking and status monitoring, role-based access control, and recruitment reports. It replaces a fragmented recruitment workflow in which vacancy notices, submitted requirements, applicant records, qualification review, and status updates are handled through separate manual or semi-digital methods.



The project uses the Waterfall model of the Software Development Life Cycle. In this study, each phase produces a defined output before the next phase begins: requirements are documented before design; design artifacts guide development; testing follows implementation; and deployment follows testing. The

documented project plan uses this sequential approach because the system requirements were identified at the early stage of development

The system design uses the **Waterfall Model** of the Software Development Life Cycle. Waterfall is a sequential design approach: requirements analysis is completed first; the approved requirements guide the system design; the design guides development; testing verifies the implemented functions; and deployment prepares the completed portal for use. The model fits this study because the portal features, user roles, quality characteristics, and performance measures are defined before development and must be traceable from requirements to design, testing, and evaluation.

In the requirements-analysis stage, the study identifies the needs of applicants, HR personnel, and authorized administrators. The system design stage converts these needs into a three-tier architecture, process models, database relationships, user interfaces, qualification criteria, role permissions, and test cases. The development stage implements the six approved features. The testing stage verifies that records process correctly, users access only their authorized functions, and the screening workflow produces a result HR personnel can review. Deployment prepares the browser-accessible system, documentation, and a controlled operational environment.

The Waterfall design does not give the scoring module final hiring authority. The system compares applicant information with the published vacancy criteria

and presents the score and its basis for HR review. HR personnel verify the complete application, apply institutional recruitment procedures, and make the final hiring decision. The completed portal is evaluated through functional suitability, compatibility, performance efficiency, interaction capability, reliability, security, maintainability, flexibility, safety, candidate-screening time, and HR-validated automated qualification-scoring accuracy.

Research and Development Methodology

The study uses a research-and-development approach. Requirements gathered from the HR recruitment process are translated into a system design, developed as a working portal, tested, and evaluated against the study objectives.

Research and Development Methodology Justification

A research-and-development approach is appropriate because the study does not only describe a recruitment problem; it produces a Human Resource Department Web-Based Job Portal System as the research output. The approach begins by determining the functional and non-functional requirements, then converts these requirements into a designed and implemented system. It ends by evaluating the completed portal through the nine adopted ISO/IEC 25010:2023 characteristics, candidate-screening time, and HR-validated automated qualification-scoring accuracy. This sequence directly answers the study's research questions and specific objectives

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

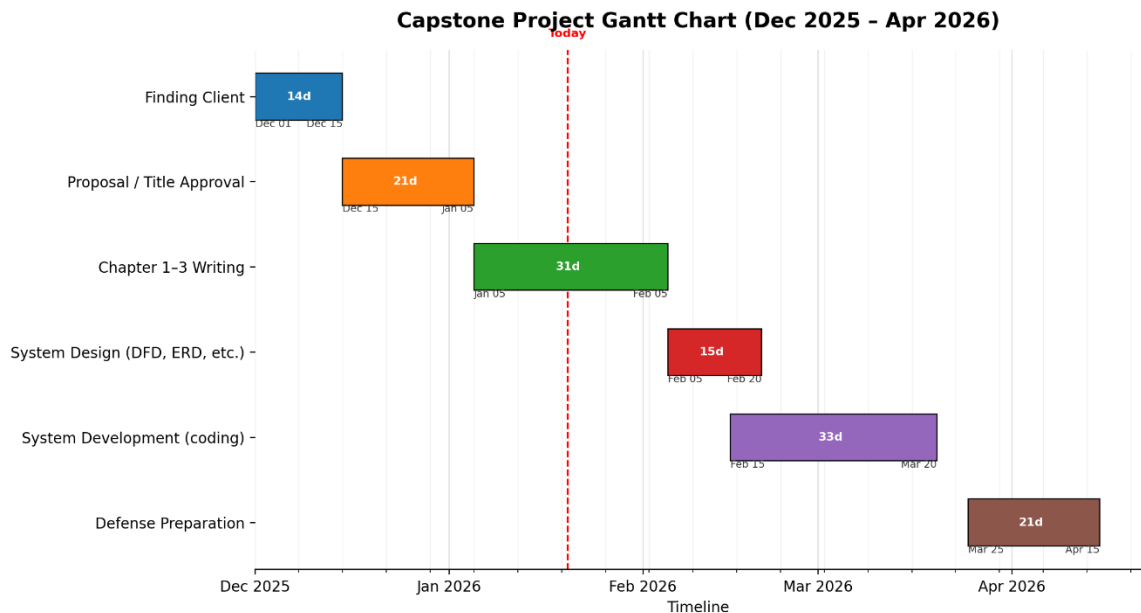


Figure 2. Gantt Chart

The Gantt chart illustrates the timeline of activities conducted in the development of the Human Resource Department Web-Based Job Portal System for NAAP, starting from December 2025 to April 2026. It shows the sequence and duration of each phase of the project to ensure proper planning and organization of tasks. The first phase is Finding Client, which was done in December 2025 to identify the project stakeholders and gather initial requirements. This was followed by Proposal or Title Approval, also conducted in December 2025, to formally approve the study and define the scope of the system. The Chapter 1 to 3 Writing phase was done from January to mid-February 2026, which included the introduction, methodology, and system design of the study. At the same time, the System Design (DFD, ERD, etc.) was completed in February 2026 to prepare the

structure of the system. The System Development (coding) phase took place from mid-February to March 2026, where the actual system was built and implemented using web technologies. Finally, Defense Preparation was conducted in April 2026 to finalize all documents, review the system, and prepare for presentation and defense.

Waterfall Development

The development of the Human Resource Department Web-Based Job Portal System for the National Aviation Academy of the Philippines (NAAP) followed the Waterfall Model of the Software Development Life Cycle (SDLC). This model was selected because it provides a structured and sequential approach, ensuring that each phase of the system is completed and validated before proceeding to the next stage. This approach is suitable for the project since the system requirements were clearly identified during the early stages of development.

Requirements Analysis

In this stage, detailed information about the system's needs was gathered and analyzed. The researchers carefully studied how HR personnel handle job postings and applicant data, as well as how applicants interact with job opportunities. From this, specific system requirements were identified. For example, HR administrators needed features for posting jobs, reviewing

applicants, and generating reports, while applicants required functionalities such as registration, job browsing, and application submission. Additionally, important considerations such as system security, data privacy, and ease of use were also defined. This phase ensured that all expected features and system behaviors were clearly documented before development began.

System Design

Once the requirements were fully understood, the System Design Phase followed. The researchers translated the identified requirements into a structured system design. They created diagrams such as Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), and system architecture models to visualize how the system would function. The design also included the layout of the user interface for both administrators and applicants to ensure a user-friendly experience. Decisions regarding technologies were finalized, including the use of HTML, CSS, and JavaScript for the frontend, PHP for backend processing, and MySQL for database management. This phase acted as a blueprint, guiding the developers on how to build the system correctly and efficiently.

Development

With a clear design in place, the development phase began, where the actual system was built. During this stage, the researchers started coding the system based on the planned design. The frontend interface was developed to allow users to interact with the system, while the backend handled data processing, authentication, and system logic. Key features such as user registration and login, job posting management, applicant tracking, and document uploading were implemented. The database was also structured to store and organize all necessary information, including user accounts, job listings, and application records. Throughout this phase, different modules were gradually integrated until a fully functional system was achieved.

Testing

After development, the system underwent the Testing Phase to ensure its quality and reliability. The researchers carefully tested each feature to verify that it worked as intended. For instance, login functionalities were checked for proper authentication, job application processes were tested to ensure data was correctly stored, and system responses were evaluated for accuracy. Any errors or issues encountered during testing were identified and corrected. The system was also evaluated in terms of usability to ensure that both HR personnel and applicants could easily navigate and use the platform. This phase was essential in improving the system's performance and ensuring that it met the project requirements.

Deployment

Finally, the researchers proceeded to the Deployment Phase, where the completed system was prepared for actual use and presentation. The system was set up in a web server environment, allowing it to be accessed through a browser. Final adjustments and improvements were made to ensure that the system was stable and ready for demonstration. The researchers also prepared the necessary documentation and presentation materials for the project defense. At this stage, the system was fully operational and capable of supporting the recruitment processes of the Human Resource Department.

Maintenance

System Development Framework

The System Development Framework illustrates the overall structure of how data moves through the Human Resource Department Web-Based Job Portal System, following an Input-Process-Output (IPO) model. The Input stage consists of job vacancy details entered by HR administrators and application requirements, resumes, and credentials submitted by applicants. The Process stage covers the

core operations of the system, including applicant registration, application submission, automated screening and scoring against job qualification criteria, and status updates. The Output stage produces results such as ranked candidate lists, match score breakdowns, updated application statuses, and real-time notifications delivered to both HR personnel and applicants.

This framework ensured every module was designed around a clear, traceable data flow from initial submission to the final hiring decision.

Current Evaluation Procedures

The Human Resource Department of the National Aviation Academy of the Philippines (NAAP) currently conducts its recruitment process primarily through manual and semi-digital methods. Job vacancies are typically announced through bulletin boards, emails, or external job platforms, while applicants submit their requirements through email or physical documents. HR personnel manually review submitted applications, organize applicant records, and track hiring progress using spreadsheets or paper-based files.

HR also communicates with applicants about interview schedules and application updates through email or phone calls. While these procedures allow the recruitment process to function, they require significant time and effort from HR staff. Evaluation of applicants is conducted by reviewing resumes and documents

individually, which may lead to inconsistencies in decision-making due to the absence of a centralized and standardized system. The lack of automation limits HR personnel's ability to filter, sort, and analyze applicant data efficiently.

Limitation and Problem Analysis

Despite its functionality, the existing system presents several limitations. One major issue is the lack of a centralized platform, which results in scattered data across different files and communication channels. This makes it difficult to track applicant progress and retrieve information quickly.

Another problem is the time-consuming nature of manual processing. HR personnel must manually review applications, organize documents, and update applicant status, which increases workload and delays the recruitment process.

The system is also prone to human errors, such as misplaced documents, duplicate entries, and incorrect data recording. Additionally, accessibility is limited, as applicants cannot easily track their application status, and HR staff cannot access data efficiently from different locations.

Furthermore, the current process lacks real-time communication and notifications, leading to delays in informing applicants about updates. Security is also a concern, as sensitive applicant data may not be fully protected in manual or loosely managed systems.

Requirement Gap Analysis

To identify the specific improvements required, the researchers compared the capabilities of the current manual process against the capabilities HR personnel actually need to operate efficiently. This comparison revealed four major gaps that the proposed system must address.

In terms of data centralization, the current process relies on scattered spreadsheets, physical folders, and email threads. At the same time, HR requires a single centralized database that consolidates all applicant and vacancy records for immediate retrieval.

In terms of process automation, applicant screening and status updates are currently performed manually by HR staff, whereas the required capability is an automated tracking mechanism that filters, sorts, and updates applicant records with minimal manual intervention.

In terms of data security and applicant visibility, the current setup lacks controlled access and real-time status tracking. At the same time, the required system must provide role-based access control, encrypted storage of applicant records, and a self-service portal where applicants can monitor their application progress without contacting HR directly.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Current State Limitation	System Requirement	Gap Resolved By
Inefficient Manual Screening & Context-Switching: HR personnel spend excessive time manually reviewing physical resumes and specialized aviation credentials (CAAP ratings, Civil Service eligibility, PDS entries), causing workflow delays and evaluator fatigue.	Automated, objective candidate parsing and qualification match scoring aligned with NAAP job specifications.	Automated PDS Screening & Scoring Engine: Laravel backend logic that parses structured digital Personal Data Sheet (PDS) submissions, evaluates criteria (education, eligibility, experience, training hours), and generates an objective 0–100% suitability match score.
Fragmented Records & Data Duplication: Applicant data is scattered across physical filing cabinets, loose email threads, and unsynchronized spreadsheets, causing misplaced documentation and zero centralized visibility.	A centralized relational database architecture for real-time record synchronization and instant candidate retrieval.	Centralized MySQL Database & Admin Dashboard: Consolidated data repository enforcing relational foreign key constraints, paired with React UI search and filtering optimization.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Lack of Applicant Visibility & Communication Overhead: Job seekers have zero real-time visibility into their application status, resulting in high volumes of repetitive administrative inquiries via phone and email.	A self-service applicant portal providing real-time status tracking and transparent communication channels.	Visual Application Timeline & Integrated Messaging: Applicant-side React portal displaying a live progress bar (Submitted → Under Review → Shortlisted → Interview → Hired/Rejected) and two-way messaging.
Unsecured Personal Records & Privacy Vulnerabilities: Physical PDS filings and unencrypted digital logs expose sensitive personal credentials to unauthorized viewing and data privacy breaches.	Controlled access, encrypted data storage, and strict compliance with the Data Privacy Act of 2012 (RA 10173).	The security architecture incorporates Role-Based Access Control to restrict data boundaries, Bcrypt for password hashing, session encryption, and server-side input validation.

Alternative Solution and Feasibility Analysis

1. **Alternative A (Enhancement of Existing Manual System)** - because it does not fully eliminate the core problems of the system. The process remains manual, making it prone to human errors such as duplicate entries and misplaced records. It also fails to provide centralized data management, real-time access, and automation, which are necessary to improve efficiency and decision-making in recruitment.

2. **Alternative B (Use of Third-Party Recruitment Platforms)** - because most third-party platforms are designed for general use and may not align with the specific workflow and requirements of NAAP's Human Resource Department. In addition, these systems often require subscription fees, offer limited customization, and may pose concerns regarding data privacy and control over sensitive applicant information.

3. **Alternative C (Custom Web-Based Job Portal System – Selected Solution)** - because it directly addresses the identified limitations of the existing system. It eliminates scattered data by using a centralized database, reduces manual workload through automation, minimizes human errors, and improves accessibility through a web-based platform.

Additionally, it enhances communication by providing real-time notifications and ensures data security through authentication and role-based access control.

Proposed Technology Alternatives

The proposed system introduces specific technological solutions to replace the inefficiencies of the manual recruitment process and address the identified system gaps:

- 1. Centralized Database Management System**

A database-driven solution that consolidates all applicant records, job postings, and recruitment data into a single platform. This eliminates scattered files and enables real-time data access, improving tracking, retrieval, and overall data consistency.

- 2. Automated Application Tracking and Notification System**

A system module that automatically updates application statuses and sends notifications to applicants regarding their progress (e.g., application received, under review, accepted, or rejected). This reduces manual communication efforts and improves transparency between HR personnel and applicants.

3. Search and Filter Optimization for Applicant Management

An integrated search and filtering feature that allows HR personnel to quickly locate applicants based on qualifications, job position, or application status. This enhances decision-making speed and reduces the time spent reviewing large volumes of applications.

Feasibility Study

Technical Feasibility

The proposed system is technically feasible as it utilizes widely available and reliable web technologies such as HTML, CSS, JavaScript, PHP, and MySQL. These tools are compatible with standard computer systems and do not require specialized hardware. The system follows a web-based client-server architecture, allowing users to access the platform through a browser. It can be deployed using common development environments such as XAMPP, making implementation practical and achievable. Additionally, the researchers possess the necessary technical knowledge and skills to develop and maintain the system successfully.

Operational Feasibility

The system is operationally feasible as it is designed to align with the workflow of the Human Resource Department. The interface is user-friendly and structured to simplify recruitment tasks such as job posting, applicant review, and

report generation. By automating repetitive processes, the system reduces the workload of HR personnel while maintaining a familiar workflow. Applicants also benefit from an intuitive interface that allows easy job searching, application submission, and status tracking. Minimal training is required, making the system practical for daily use.

Economic Feasibility

The project is economically feasible because it utilizes open-source technologies, eliminating the need for expensive software licenses. Development is conducted within an academic setting, which further minimizes initial costs. In the long term, the system reduces operational expenses by minimizing paper usage, decreasing manual labor, and improving efficiency. Faster recruitment processes and better data management also contribute to indirect cost savings for the organization.

Schedule Feasibility

The project is schedule feasible as it follows a structured timeline aligned with the academic calendar. Based on the Gantt Chart, the development phases—from planning, analysis, design, development, testing, and deployment—were completed within the period of December 2025 to April 2026. The use of the Waterfall Model ensured that each phase was completed sequentially and on time.

Proper time allocation and task management allowed the researchers to develop, test, and finalize the system within the given timeframe.

Cost Benefit

Overview

The cost–benefit analysis evaluates the financial viability of the NAAP HR Job Portal by comparing projected development and operational costs against tangible and intangible benefits over a three-year horizon. This analysis ensures that the investment in digital transformation is justified not only by direct monetary savings but also by institutional improvements in efficiency, transparency, and compliance.

Cost Analysis

The cost analysis includes the labor required to build the system following the Waterfall SDLC and the long-term operational expenses. Labor rates are calculated based on professional standards, including ₱550/hour for Developers and Project Managers.

Development Costs

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Development costs are calculated based on labor rates and hours required for each phase of the Waterfall SDLC.

Role	Hourly Rate (₱)
Project Manager	2500
Developer / Programmer	2500
System Analyst	2000
UI/UX Designer	2000
QA / Test Specialist	1500

Phase 1: Requirement Analysis

This phase involves gathering specific operational needs for the NAAP HR Department, focusing on transitioning from manual resume handling to digital tracking.

Activity	Role	Hours	Computation	Cost (₱)
Requirement Gathering (HR Workflow)	System Analyst	8	1 x 550 x 8	4,400
Project Scoping and Planning	Project Manager	4	1 x 550 x 4	2,200
HR Stakeholder Consultation	Full-Stack Developer	6	1 x 550 x 6	3,300
Workflow Documentation	QA Specialist	5	1 x 400 x 5	2,000

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Documentation Materials	Project Manager / Developer	-	Estimated	50
Total Cost				11,550

Phase 2: System Design

Translating HR requirements into a technical blueprint, including the database schema for applicant records and vacancy details.

Activity	Role	Computation	Cost (₱)
Architecture Supervision	Project Manager	1 x 550 x 8	4,400
System Logic and DFD	System Analyst	1 x 550 x 10	5,500
Database Schema and ERD	Developer	1 x 550 x 12	6,600
Portal Wireframing	UI/UX Designer	1 x 400 x 15	6000
Total cost			22,500

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Phase 3: Implementation (Coding)

Developing the React.js/Inertia frontend and Laravel backend, integrating the role-based access control for HR and Applicants.

Activity	Role	Hours	Computation	Cost (₱)
Full-Stack Portal Development	Developer	45	1 x 550 x 45	24,750
Logic and RBAC Structuring	System Analyst	16	1 x 550 x 16	8,800
Development Supervision	Project Manager	7	1 x 550 x 7	3,850
HR Dashboard Integration	UI/UX Designer	8	1 x 400 x 8	3,200
Development Utilities	Internet/Cloud	-	Estimated	4,350
Total cost				44,950

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Phase 4: Testing

Executing functional tests to ensure resume uploads and application status tracking work correctly across NAAP campuses.

Activity	Role	Hours	Computation	Cost (₱)
Test Plan and Execution	QA Specialist	20	1 x 350 x 20	7,000
Bug Fixing and Logic Adjustment	Developer	10	1 x 550 x 10	5,500
UAT Coordination	QA Specialist	5	1 x 350 x 5	1,750
User Acceptance (HR Dept)	Project Manager	5	1 x 550 x 5	2,750
Total Cost				17,000

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Phase 5: Deployment

Migrating legacy recruitment logs to the centralized MySQL database and training NAAP HR staff.

Activity	Role	Hours	Computation	Cost (₱)
Deployment Planning	Project Manager	5	1 x 550 x 5	2,750
Server Setup and Data Migration	Developer	9	1 x 550 x 9	4,950
HR Training Preparation	System Analyst	5	1 x 550 x 5	2,750
HR Portal Walkthrough	Project Manager	5	1 x 550 x 5	2,750
Security and Web Hosting (1 Yr)	Hosting/SSL	-	Estimated	3,481
Total Cost				16,681

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Phase 6: Maintenance

Ongoing system monitoring to ensure security updates and portal uptime.

Activity	Role	Hours	Computation	Cost (₱)
Performance Monitoring	Developer	6	1 x 550 x 6	3,300
Security Issue Diagnosis	QA Specialist	6	1 x 350 x 6	2,100
Documentation Updates	System Analyst	4	1 x 550 x 4	2,200
Total Cost				7,600

Total Development Cost

Phase	Description	Total Cost (₱)
Phase 1	Requirement Analysis	11,550
Phase 2	System Design	22,500
Phase 3	Implementation (Coding)	44,950
Phase 4	Testing and Integration	17,000

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Phase	Description	Total Cost (₱)
Phase 5	Deployment	16,681
Phase 6	Maintenance	7,600
Total Project Cost	Overall Waterfall Development	120,281

3-Year Operational Cost Projection

Category	Description	Estimated Cost (₱)
Hardware	Existing NAAP Computer Setup	0
Software	SSL, Hosting, Domain, and Cloud Backups	30,000
Labor	System design and database optimization	45,000
Miscellaneous	Utilities and Documentation Materials	5,000
Total Estimated Cost	Projected Budget for 3 Years	80,000

Benefit Analysis (3 Years)

The benefits are divided into tangible savings (direct money saved) and intangible improvements (qualitative gains in efficiency).

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Tangible Benefits

Category	Description	Value (P)
Stationery	Elimination of logbooks, pens, and filing materials	320
Labor Recovery	Recovered hours from manual data entry and tracking	48,000
Total Tangible (Annual)	Yearly Savings	48,320
3-Year Total Benefits	Annual Total × 3 Years	144,960

Intangible Benefits

- Improved Accuracy: Minimizes human error in record-keeping and data entry.
- Enhanced Decision-Making: Analytics provide real-time insight into recruitment trends.
- Data Security: Secure web-based architecture ensures real-time monitoring and cloud backups.
- Institutional Excellence: Modernizes the recruitment process to meet professional digital standards.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Total-Benefit Summary

This summary provides the project's final financial health, showing that the savings significantly outweigh the costs.

Description	Amount (₱)
Total Project Cost (3 Years)	80,000
Tangible Benefits (3 Years)	144,960
Net Benefits (3 Years)	64,960
Return on Investment (ROI)	81.2%
Benefit-Cost Ratio (BCR)	1.81:1

Conclusion

The 3-Year Cost-Benefit Analysis confirms that the **Human Resource Department Web-Based Job Portal System** is economically feasible and highly beneficial for the **National Aviation Academy of the Philippines (NAAP)**. With a total project cost of ₱80,000—covering all initial development, software, and three years of maintenance—the system is projected to yield ₱144,960 in tangible savings over the same period. This results in a Net Total Benefit of ₱64,960 and a positive Benefit-Cost Ratio (BCR) of 1.81:1, indicating a solid and secure return on investment for the institution.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Beyond the direct financial returns derived from preventing the use of manual logs and reducing administrative overhead, the system provides crucial institutional value. The transition from a slow, paper-based recruitment process to an automated digital platform modernizes NAAP's operations and ensures that applicant tracking is consistent across all campuses. **The system mitigates critical operational risks such as lost applicant data or duplicated records and greatly improves the daily management efficiency of the HR department.**

Ultimately, the investment in this custom digital solution allows NAAP to shift from a reactive **manual process to a proactive, data-driven operation. By automating job postings, resume storage, and status notifications, the academy can focus its resources on strategic talent acquisition rather than time-consuming paperwork.** This digital transformation ensures the recruitment process remains **stable, secure, and ready to scale alongside the academy's future growth.**

Cost Benefit summary

This summary provides the final financial health of the project, demonstrating that the savings significantly outweigh the costs.

System Requirements Specification

The **System Requirements Specification (SRS)** for the **Human Resource Department Web-Based Job Portal System of the National Aviation Academy of the Philippines (NAAP)** is designed to transform the existing manual recruitment process into a **centralized, web-based platform**. The system aims to improve the **efficiency of handling job applications by providing a digital environment where applicants can submit requirements online, and HR personnel can manage recruitment processes in an organized manner**.

By integrating job posting, applicant management, and application tracking into a single system, all recruitment-related data is stored and processed digitally. This eliminates the limitations of manual processing, such as data redundancy, delays, and difficulty in tracking applicants. Furthermore, the system provides reporting features and automated qualification match scoring that assist the HR Department in monitoring application progress and improving decision-making.

Stakeholder Identification and User Roles

The primary stakeholders of the system are the Human Resource (HR) Department of NAAP and the job applicants. The **HR Department** serves as the **main user and decision-maker, responsible for managing job postings, reviewing applications, evaluating applicant qualifications, and monitoring applicant records**. To support the recruitment workflow, the system is designed with distinct user roles.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

1.) **Admin (HR Personnel / HR Staff / Super Admin):** Has administrative control over the system, including creating and editing job vacancies, setting custom file requirements, reviewing applicant Personal Data Sheet (PDS) submissions, viewing automated AI qualification match scores (0–100%), updating application statuses (Submitted, Under Review, Shortlisted, Interview Scheduled, Hired, Rejected), scheduling interviews, communicating with applicants via integrated messaging, monitoring staffing positions, and exporting summary reports.

2.) **Applicants (Job Seekers):** Interact with the system by registering accounts, managing profile details, browsing available non-academic job openings, submitting digital application forms, uploading required documents (PDF/DOCX resumes, PDS, certifications), managing To-Follow document submissions, and tracking their application status in real time.

This role distribution ensures that both administrative and applicant-side processes are handled efficiently within a single platform.

Functional Requirements

- 1.) **User Registration and Authentication Function:** Allows applicants and administrators to securely access the system using login credentials. Features include password encryption (Bcrypt), session management, Role-Based Access Control (RBAC), and optional Two-Factor Authentication (2FA) via TOTP / Google Authenticator for administrative accounts.

- 2.) **Job Posting Management Function:** Enables HR administrators to create, update, publish, close, and archive job vacancies. Admins can specify job titles, departments, employment types, salary grades, job descriptions, responsibilities, deadline dates, and custom document upload requirements.

- 3.) **Application Submission Function:** Allows applicants to apply for open job postings, complete digital Personal Data Sheet (PDS) entries, upload required credentials (resumes, transcripts, Civil Service eligibility forms), and specify pending To-Follow documents.

- 4.) **Application Tracking Function:** Enables job applicants to monitor the real-time progress of their submitted applications through a visual status

timeline (Submitted, Under Review, Shortlisted, Interview Scheduled, Hired, Rejected).

- 5.) **Applicant Management & Qualification Scoring Function:** Enables HR personnel to view, filter, sort, and evaluate applicant data. The system features an automated PDS Qualification Match Scoring Engine that calculates an objective suitability score (0–100%) based on Education Level, Years of Experience, Awards/Eligibility, Training Hours, and Document Completeness to accelerate candidate shortlisting.
- 6.) **Messaging and Interview Scheduling Function:** Supports direct two-way messaging between HR administrators and applicants regarding application updates, document requests, or interview details. Includes an integrated calendar system for scheduling and tracking candidate interview appointments.
- 7.) **Report Generation Function:** Generates downloadable summary reports (CSV/Excel) of recruitment metrics, applicant lists, vacancy distributions, and application status statistics to support data-driven decision-making.

- 8.) **Activity & Audit Logging Function:** Records administrative actions and system events into security audit logs to preserve data transparency and institutional auditability.

Non-Functional Requirements

- 1.) **Security:** Ensures sensitive user and applicant data are protected through password hashing (BCrypt), input sanitization, server-side data validation, Role-Based Access Control (RBAC), session encryption, and optional 2FA. Compliance with the Philippines Data Privacy Act of 2012 (RA 10173) safeguards personal credentials and PDS filings.
- 2.) **Usability:** Provides a clean, intuitive, and accessible user interface built using modern UI standards (Tailwind CSS, Radix UI). The interface ensures learnability and visual clarity for both HR personnel and job applicants across desktop, tablet, and mobile viewports.
- 3.) **Performance Efficiency:** Ensures rapid client-side page transitions and quick data retrieval within < 1.5 seconds, utilizing Inertia.js single-page application (SPA) routing, Vite asset optimization, and indexed database queries.

4.) **Reliability:** Maintains consistent system functionality without errors during high concurrent user activity. Database transactions enforce foreign key constraints and cascading integrity to prevent corrupted or orphaned records.

5.) **Maintainability & Scalability:** Designed using a modular Model-View-Controller (MVC) component architecture (Laravel 11 backend and React 19 frontend), allowing future expansion, updates, and feature enhancements with minimal system refactoring.

Proposed System Architecture and Design

The proposed system is designed to support a web-based recruitment process using a structured and organized architecture. It follows a three-tier architecture, separating the system into presentation, application, and data layers. This separation ensures better performance, security, and maintainability of the system. The system focuses on providing a smooth interaction between users and the platform, allowing real-time processing of applications and data management. Unlike manual systems, this design ensures that all recruitment activities are handled digitally and efficiently.

System Architecture Overview

Adopting a web-based client-server architectural model, the system provides a centralized recruitment platform for the Human Resource Department of NAAP at the Villamor Air Base campus. This architecture enforces clear separation between the user interface, application processing logic, and data storage, ensuring the system remains modular, maintainable, and scalable.

I. Presentation Layer (Frontend System)

The Presentation Layer serves as the primary interface for both job applicants and HR administrators. Developed using **React 19**, **TypeScript**, **Inertia.js**, and **Tailwind CSS**, it provides a responsive, Single-Page Application (SPA) environment.

For Applicants: Allows account registration, job searching, digital form completion, document uploading, status timeline tracking, and HR messaging.

For HR Administrators: Provides an administrative control panel for managing job vacancies, evaluating applicant PDS match scores, reviewing

candidate documents, scheduling interviews, exporting reports, and tracking staffing positions.

II. Application Programming Interface Layer (Backend Processing)

The Application Programming Interface (API) Layer serves as the core processing engine of the system. Built on **Laravel 11 (PHP 8.2+)**, it processes all server-side logic, routing, middleware checks, and data requests transmitted from the frontend. This layer manages key operations including user authentication, session security, password encryption, input validation, candidate suitability scoring, file handling, and status progression. By separating business logic from the user interface, it guarantees data security, integrity, and smooth interaction between users and the database.

III. Data Management Layer (Database System)

The Data Management Layer handles storage, organization, and retrieval of all system data using a MySQL relational database engine. It stores user account credentials, job vacancy specifications, applicant records, PDS entries, application statuses, internal messages, interview schedules, staffing positions, and activity logs. The database enforces relational integrity through foreign key constraints, enabling real-time record synchronization for HR administrators and applicants.

System Components Modules

The system is organized into specialized modules, each designed to address gaps identified in the Human Resource Department's manual recruitment process at the National Aviation Academy of the Philippines (NAAP). The platform is developed as a web-based system using HTML, CSS, JavaScript, PHP, and MySQL, ensuring a responsive and centralized recruitment management system. The system allows real-time processing of job applications, ensuring that applicant data, job postings, and recruitment status are consistently updated and properly managed within the database.

- 1.) **User Authentication Module** – This module handles the secure login and registration of both applicants and HR administrators. It ensures that only authorized users can access the system by validating credentials and managing user sessions. This module also protects sensitive user information through proper authentication mechanisms.

- 2.) **Job Posting Management Module** – This module allows the HR administrator to create, update, and delete job vacancies. It ensures that all job postings are accurately displayed to applicants in real time, providing updated employment opportunities within the institution.

- 3.) **Application Submission Module** – This module enables applicants to apply for available job postings by filling out application forms and uploading required documents such as resumes and credentials. All submitted applications are stored directly in the database for proper record management.
- 4.) **Application Status Tracking Module** – This module allows applicants to monitor the progress of their job applications. It provides real-time updates such as pending, reviewed, shortlisted, or approved status, ensuring transparency throughout the recruitment process.
- 5.) **Applicant Management Module** – The HR administrator uses this module to view, organize, and evaluate applicant data. It allows filtering and reviewing of applications to support efficient decision-making during recruitment.
- 6.) **Admin Dashboard Module** – This module serves as the central control panel of the system. It provides an overview of job postings, applicant submissions, and recruitment activities, allowing the HR administrator to manage and monitor the entire system efficiently.

7.) **Report Generation Module** – This module generates summarized reports of recruitment activities such as the number of applicants per job posting and overall application status distribution. It helps the HR Department analyze recruitment performance and improve decision-making.

8.) **Database Management Module** – This module is responsible for storing and managing all system data, including user accounts, job postings, application records, and uploaded documents. It ensures data integrity, security, and consistency across the entire system.

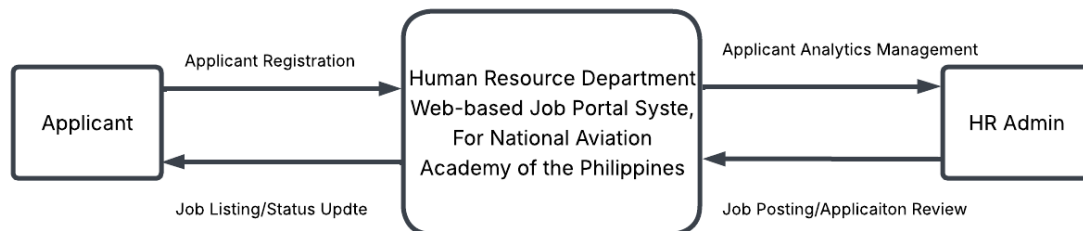


Figure 3. Context Diagram

The context diagram of the Human Resource Department Web-Based Job Portal System presents a high-level view of how the system interacts with its external entities. It shows the overall flow of data between the users and the system, without going into detailed processes. At the center of the diagram is the

HR Job Portal System, which serves as the main process that handles all inputs and outputs. Connected to the system are two main external entities: the Applicant and the Admin.

The applicant interacts with the system by submitting personal information, applying for jobs, and uploading required documents. In return, the system provides outputs such as job listings, application status updates, and notifications.

On the other hand, the admin interacts with the system by managing job postings, reviewing applications, and monitoring applicant activity. The admin also receives outputs such as reports and system updates that support decision-making.

All interactions are made possible through the internet, which acts as the communication medium between users and the system. The system processes and stores all data in a centralized database, ensuring that information is organized, secure, and easily accessible.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

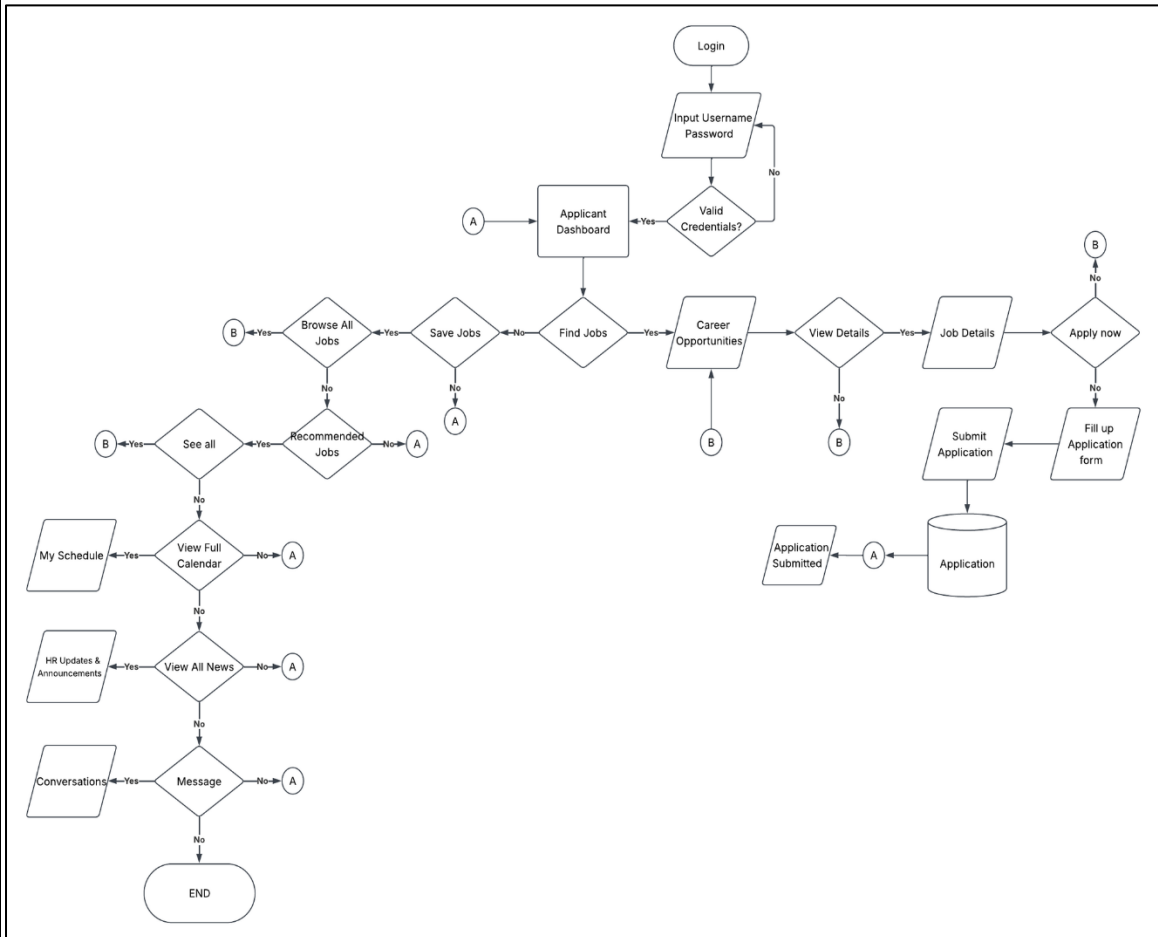
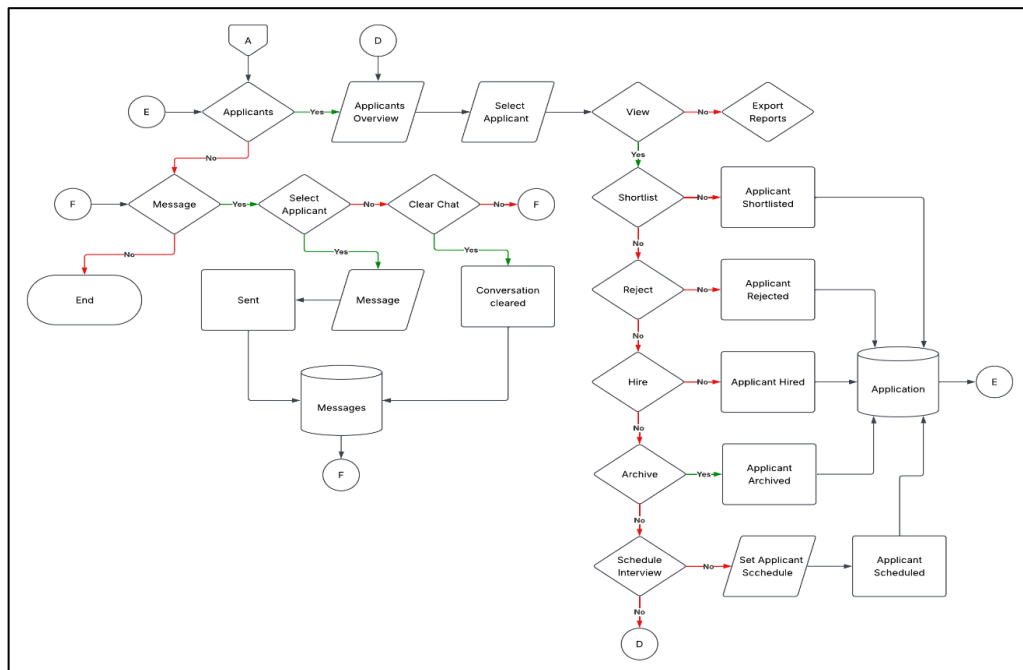
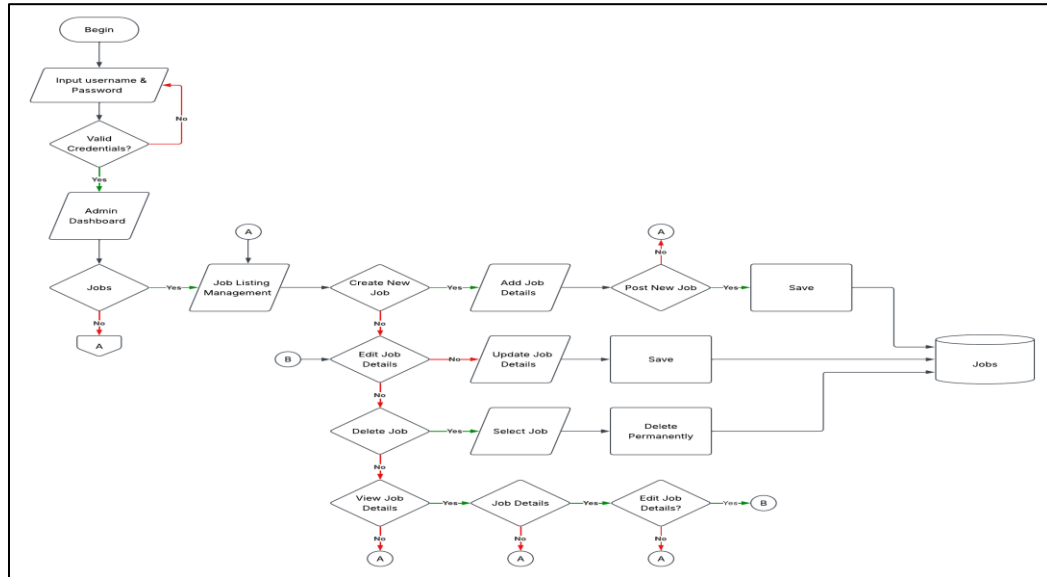


Figure 4 Applicant Flow Chart

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES



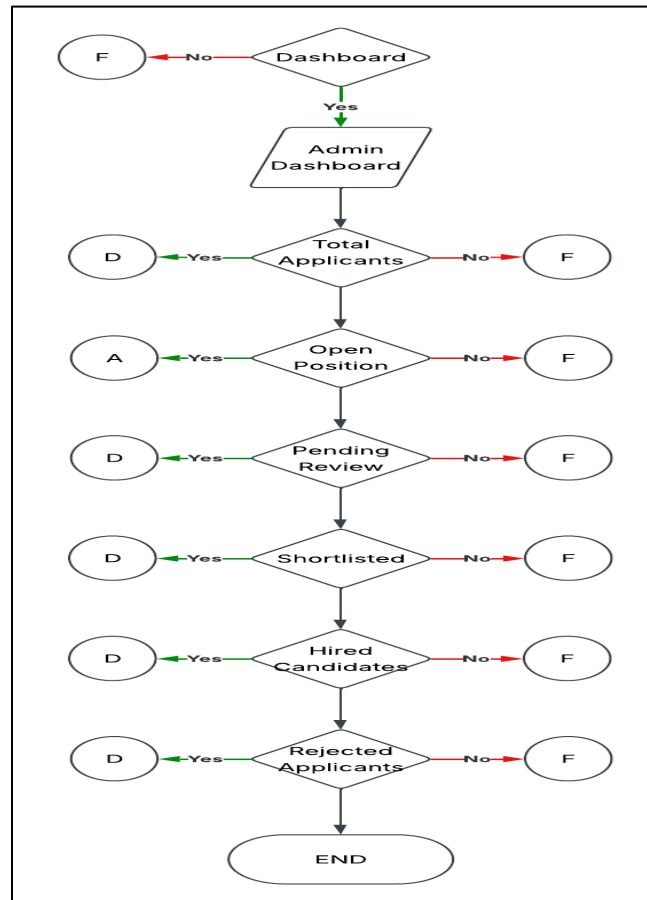


Figure 5. Applicant & Admin Flowchart

The system flowchart illustrates the overall process of the web-based job application and management system, integrating both administrator and applicant

functionalities into a single unified workflow. This flow ensures a seamless interaction between users and the system while maintaining security, data integrity, and efficient processing.

The process begins at the system start, where users access the web application through a browser. Upon loading, the system directs users to the authentication page, where they are required to either log in or register. New users must complete the registration process by providing necessary personal information, which is validated before account creation. Existing users enter their credentials, and the system verifies them to grant access.

Once authenticated, the system determines the user role (administrator or applicant) and redirects the user to the appropriate interface.

For administrators, the system provides access to the main dashboard, where they can manage core functionalities. Administrators can perform job management, including creating, editing, deleting, and publishing job postings.

They can also handle user management, where they view, add, or update user accounts. Additionally, administrators can access applicant management, allowing them to review submitted applications, filter candidates, shortlist applicants, and manage interview schedules. The system also supports report generation and analytics, enabling administrators to monitor system performance and recruitment data. All actions performed are validated and stored in the centralized database.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

For applicants (users), the system provides access to job-related functionalities. Users can browse available job listings through the job exploration module, where search and filter features help them find relevant opportunities. Once a suitable job is identified, users can proceed to the application process, submitting required documents such as resumes and cover letters. The system validates all submitted data before storing it in the database. Users can also manage their personal information through the profile module, ensuring that their details are up to date.

Throughout the process, the system generates notifications and status updates, informing users about application progress, such as submission confirmation, interview schedules, or hiring decisions (accepted or rejected). These updates enhance communication between the system and its users.

All data transactions, including user information, job postings, and applications, are handled through a centralized MySQL database, ensuring consistency, security, and efficient retrieval of information.

The flow concludes when users choose to log out, securely ending their session. If not, they may continue navigating the system as needed.

Overall, the unified system flowchart demonstrates a complete and integrated workflow that connects authentication, role-based access, job management, application processing, and feedback mechanisms within a single

web application. This design ensures a smooth, efficient, and user-friendly experience for both administrators and applicants.

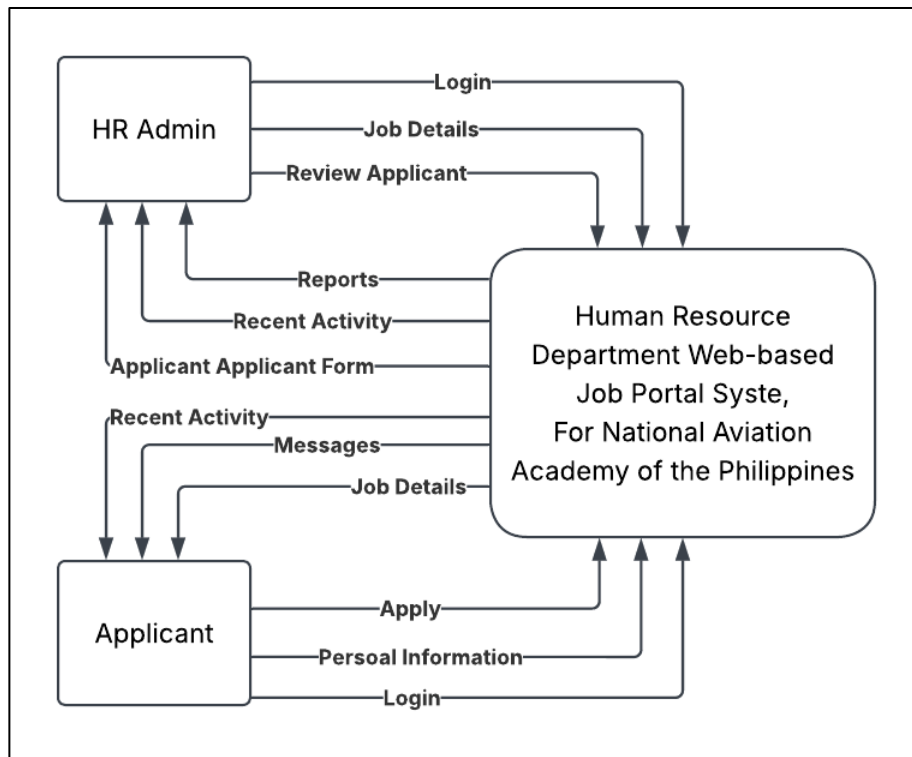


Figure 6. Data Flow Diagram Level 0

The Data Flow Diagram (Level 0) illustrated the overall flow of data within the proposed system, the HR Job Portal System. At this level, the system was represented as a single process, while its interaction with external entities was emphasized.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

The **Admin** entity was responsible for managing and maintaining the system. The admin handled job postings, review actions, and reports, ensuring that the platform functioned properly. Through the system, the admin could manage job data, evaluate applicants, and monitor overall system performance.

The **Applicant** entity interacted with the system by logging in, providing personal information, and accessing job-related data. Applicants used the system to browse job listings, submit applications, and receive updates regarding their application status.

At the center of the diagram, the **HR Job Portal System** served as the main process that handled all inputs and outputs. It processed user credentials, managed job postings, handled application data, and ensured proper data storage and retrieval.

The system also interacted with an external Email Notification Service that delivered status updates and notifications to users.

The outputs of the system included job listings, application status updates, reports, and notifications, which were provided to both administrators and applicants. This context diagram showed how the system functioned as a central

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

hub, connecting users and external services to ensure efficient job application and management processes.

Overall, the context diagram provided a high-level overview of how the system interacts with external entities such as applicants and administrators, showing only the general flow of inputs and outputs. In contrast, the Data Flow Diagram (DFD) Level 1 presents a more detailed representation of how data moves within the system by breaking it down into major processes, data stores, and interactions.

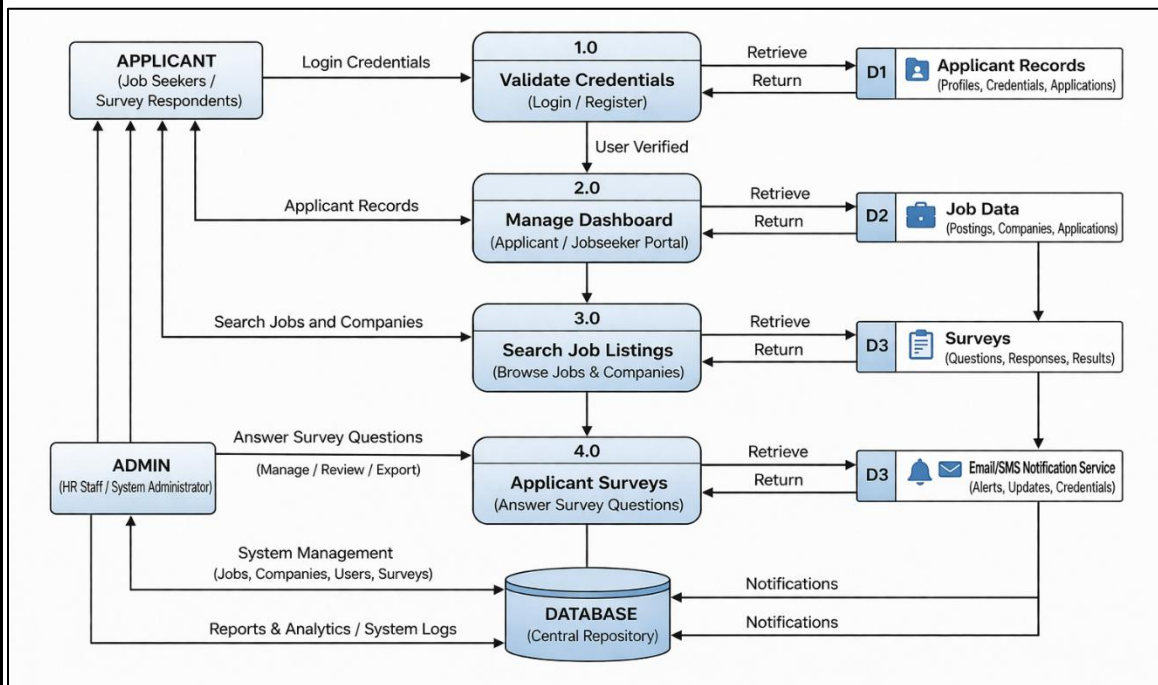


Figure 7. Data Flow Diagram Level 1

External Entities

The DFD level 1 includes two main external entities:

1.) **Applicant** – Represents job seekers who use the system to register or log in, search for available job opportunities, and participate in surveys. Applicants interact with the portal primarily through the authentication process, the dashboard interface, the job search module, and the survey system. Their actions generate data that is stored in the Applicant Records, Job Data, and Surveys data stores, ensuring that their recruitment journey is fully documented and traceable.

2.) **Admin** – Represents the Human Resource staff or system administrator responsible for managing applicants, job postings, and survey data. The admin oversees the integrity of the system by validating records, monitoring dashboards, reviewing survey responses, and ensuring that notifications are properly sent. Their role is critical in maintaining the accuracy, security, and efficiency of the recruitment process.

Process

Process 1: Validate Credentials (1.0)

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

This process handles user authentication. Applicants and admins enter their login credentials, which are verified against stored records in the Applicant Records database. Once validated, access to the system is granted. Invalid credentials trigger error messages and prevent unauthorized access, thereby safeguarding sensitive recruitment information.

Process 2: Manage Dashboard (2.0)

After successful login, users are directed to the dashboard. For applicants, the dashboard provides access to their records, submitted applications, and current application status. For admins, the dashboard functions as a control panel where they can monitor and manage system data such as user accounts, job postings, and surveys. This dual functionality ensures that both applicants and administrators can efficiently perform their respective tasks.

Process 3: Search Job Listings (3.0)

This process allows applicants to search for available job opportunities and companies. The system retrieves relevant data from the Job Data store and displays results based on the applicant's search criteria. By enabling structured queries and returning filtered results, this process supports transparency and accessibility in recruitment, ensuring that applicants can easily identify opportunities that match their qualifications.

Process 4: Search Job Listings (4.0)

Applicants answer survey questions through this process. The responses are stored in the Surveys data store and can be used for evaluation, feedback, and reporting. Notifications may also be sent via the integrated Email Notification Service, ensuring applicants remain informed about deadlines, interview schedules, and survey participation.

Data Stores

D1: Applicant Records – Stores user profiles, login credentials, and application details. This data store ensures that authentication and applicant tracking are consistent and secure.

D2: Job Data – Contains job postings, company information, and related data. It supports the job search process and provides applicants with accurate and updated vacancy information.

D3: Surveys – Stores survey questions, responses, and results. It provides HR with valuable feedback and supports continuous improvement of recruitment practices.

Database: Serves as the central repository that integrates all system data and supports notification processes. It ensures data integrity, consistency, and compliance with institutional standards.

The Data Flow Diagram (DFD) Level 1 illustrates the detailed interactions between applicants, administrators, and the central database within the NAAP HR Job Portal. It decomposes the high-level processes into specific modules, showing how data flows across the system to support recruitment and survey management.

The process begins with Validate Credentials (1.0), where both applicants and administrators input their login information. These credentials are checked against the Applicant Records (D1) data store, which contains user profiles, authentication tokens, and application histories. This ensures secure access control and prevents unauthorized entry. Successful authentication grants the user access to the portal, while failed attempts trigger error messages and optional password recovery workflows.

Once authenticated, users proceed to Manage Dashboard (2.0), which serves as the central interface of the system. For applicants, the dashboard provides access to their personal profile, submitted applications, and real-time status updates. For administrators, the dashboard functions as a control panel. It allows HR staff to manage job postings, oversee applicant records, and configure surveys. This dual functionality ensures that both user groups interact with the system in a role-appropriate manner.

From the dashboard, applicants can initiate the Search Job Listings (3.0) process. This module enables browsing of available vacancies and company information. Queries are processed against the Job Data (D2) store, which contains structured information on open positions, qualifications, and deadlines. The system returns filtered results based on applicant preferences, ensuring efficient navigation of opportunities.

Applicants also interact with the Applicant Surveys (4.0), where they answer survey questions related to employment, feedback, or system usability. Survey data is retrieved from and stored in the Surveys (D3) data store. This process not only captures applicant feedback but also supports continuous improvement of HR practices. Additionally, the survey module integrates with the Email Notification

Service, which sends alerts, confirmations, and reminders to applicants. Notifications ensure that users remain informed about deadlines, interview schedules, and survey participation.

Administrators play a critical role in maintaining system integrity. They access applicant records, review survey responses, and manage overall system operations. Through the dashboard, admins can post and update job vacancies, review applicant submissions and qualification scores, export survey data for analysis, trigger notifications to applicants, and monitor system logs. Their interaction with the central Database guarantees that all records remain consistent, secure, and auditable. This oversight ensures that recruitment processes align with institutional standards and that applicant data is handled ethically and transparently.

In summary, this **DFD Level 1** reflects the actual recruitment workflow observed in **NAAP's HR Department**, where applicants previously submitted paper resumes and HR staff manually tracked applications. **By mapping these processes into the portal, the system directly addresses the inefficiencies we documented during our requirement-gathering phase.**

Data Modeling and Database Design

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

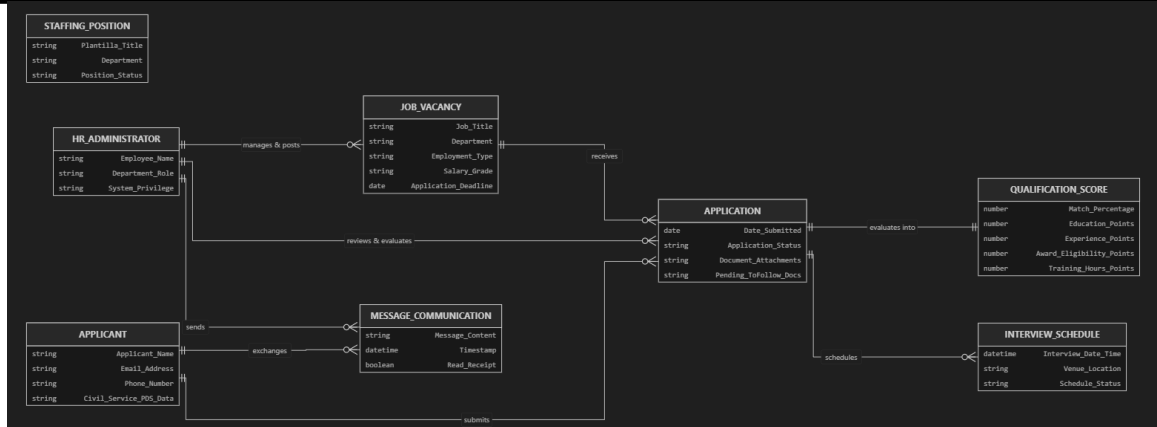


Figure 8. Conceptual Data Model

Prior to defining the technical database schema, the researchers established a conceptual data model to define the high-level domain entities, institutional business rules, and core relationships governing the NAAP HR Job Portal System. At this conceptual level, the system revolves around Applicants, who register accounts, submit Civil Service Personal Data Sheet (PDS) credentials, and file job applications; HR Administrators, who manage institutional vacancy postings, evaluate candidate qualifications, update application statuses, and track positions; and Job Vacancies, which represent open non-academic positions aligned with authorized Staffing Positions.

Furthermore, submitted Applications are evaluated by an automated scoring engine to compute an objective Qualification Score based on applicant credentials. The recruitment workflow is supported by Interview Schedules to manage candidate appointments and Message Communications to facilitate direct

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

two-way updates between applicants and HR personnel. This conceptual data model establishes the essential business entities of the National Aviation Academy of the Philippines (NAAP) recruitment pipeline and serves as the theoretical foundation for the detailed logical database design and Entity Relationship Diagram (ERD) that follow.

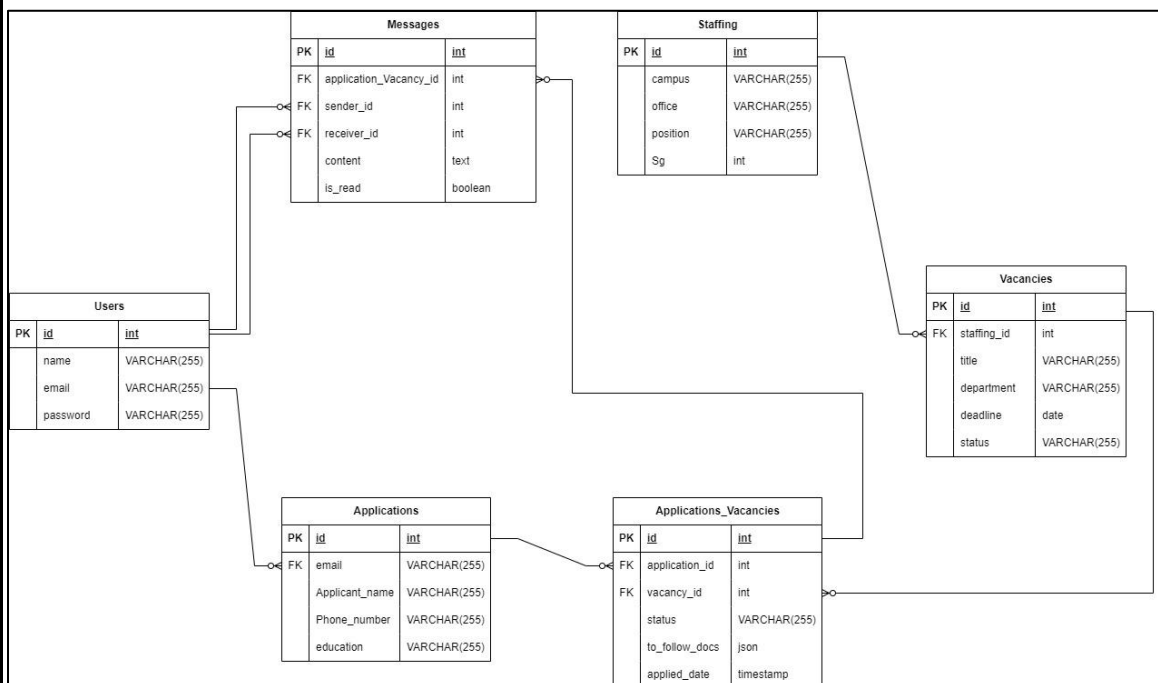


Figure 9. Entity Relationship Diagram

The **Entity Relationship Diagram (ERD)** of the Human Resource Department Web-Based Job Portal System illustrates the structure of the database

and the relationships between the system's core entities. It provides a detailed representation of how data is organized, stored, and interconnected to support the overall functionality of the system.

At the center of the diagram is the Users entity, which stores essential user information such as name, email, and password. This entity is connected to other components of the system, as users can act as applicants who submit job applications and interact with the platform.

The Applications entity contains applicant-specific information, including applicant name, contact details, and educational background. This entity represents the data submitted by users when applying for job opportunities. Each application is associated with a user and serves as a primary record of applicant information.

The Vacancies entity represents job postings created by the Human Resource administrators. It includes details such as job title, department, deadline, and status. This entity is linked to the Staffing entity, which defines organizational details such as campus, office, and position, indicating that each job vacancy is associated with a specific staffing requirement.

To manage the relationship between applicants and job vacancies, the system uses the Applications Vacancies entity. This serves as a junction table that connects applications and vacancies, enabling a many-to-many relationship. It also stores important information such as application status, submitted documents, and the date of application.

The Messages entity supports communication within the system by storing messages exchanged between users. It includes sender and receiver identifiers, message content, and read status. This entity is linked to both users and applications, allowing communication regarding specific job applications. All entities are connected through primary and foreign keys, ensuring data integrity and consistency within the database. These relationships enable efficient data retrieval, management, and processing across the system. This ERD provides the logical foundation for the NAAP HR Job Portal database, ensuring that applicant records, job postings, applications, and communication are interconnected in a secure and efficient manner. It serves as the blueprint for implementing the system's data storage and retrieval processes.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

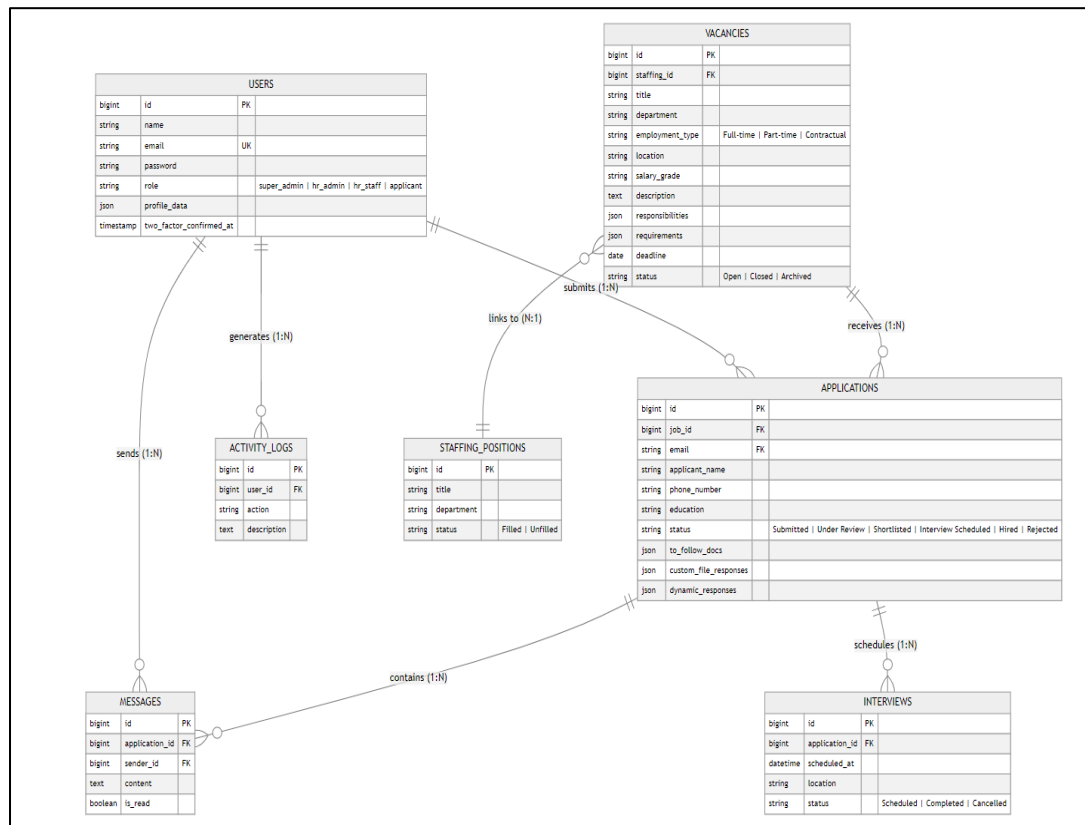


Figure 10. Logical Database Design

Building on the conceptual and entity-relationship models, the **logical database design** translates each entity into a structured table with defined fields and key relationships. The Users table stores account credentials and role information (**Applicant or HR Administrator**). The Applications table stores

applicant-specific submission details, including personal data sheet information and uploaded documents, with a foreign key reference to the Users table. The Vacancies table stores job posting details such as title, department, qualification standards, and deadline, with a foreign key linking each vacancy to its corresponding entry in the Staffing table, which tracks whether a position is filled or unfilled.

The Applications Vacancies table serves as a junction table that links Applications to Vacancies, storing the current application status and submission date, and allowing a many-to-many relationship between applicants and job postings. Finally, the Messages table stores sender and receiver identifiers along with message content and read status, enabling communication tied to a specific application. Primary and foreign key constraints across these tables maintain referential integrity and ensure consistent, accurate data throughout the system.

System Security and Data Privacy Design

Access Control and Authentication Mechanisms

The system implements **Role-Based Access Control (RBAC)** to restrict system functions according to user type. Applicants are limited to actions involving their own application records, such as submitting requirements, viewing their status, and messaging HR, while HR Administrators are granted access to manage

job postings, review ranked applicants, and update application statuses. User authentication requires a valid email address and password, with passwords stored using industry-standard hashing algorithms rather than plain text, and session handling mechanisms are used to prevent unauthorized access after periods of inactivity.

Data Encryption and Protection Measures

Sensitive applicant information, including uploaded credentials and personal data sheet details, is transmitted over encrypted HTTPS connections and stored using encrypted fields within the database. This ensures that applicant data remains protected both during transmission and while at rest, reducing the risk of unauthorized interception or data breaches.

Data Privacy and Ethical Considerations

The system's handling of applicant information adheres to the principles of the **Data Privacy Act of 2012 (Republic Act No. 10173)**, which governs the collection, processing, and storage of personal data within the Philippines. Applicants are informed of how their submitted data will be used during the application process, and access to personal records is limited strictly to authorized HR personnel. Data retention practices ensure that applicant information is kept only for as long as necessary for recruitment purposes.

Audit Logging and Activity Monitoring

To maintain accountability, the system logs key actions performed by HR Administrators, such as changes to application status, access to applicant records, and exported reports. These logs allow the academy to trace any modification back to a specific user and timestamp, supporting transparency throughout the recruitment process.

Implementation Environment and Technology Stack

Software Requirements

During development, we worked on **Windows 10 (64-bit) machines**, which gave us compatibility with **Laravel, Node.js, Composer, and MySQL**. For testing and accessing the portal, we relied on modern browsers such as **Google Chrome and Microsoft Edge**. The backend was powered by **MySQL** as our relational database, while **Apache/Nginx** served as the web server environment to host the application securely.

The system was developed using a monolithic client-server architecture composed of:

Backend:

- **Laravel PHP:** Used for backend processing. It handles business logic, authentication, and database interactions, ensuring secure and efficient server-side operations.
- **Composer:** Manages PHP dependencies. It ensures that Laravel and its required packages are installed and updated consistently across the development environment.
- **Apache or Nginx:** Acts as the web server environment to host the application. These servers deliver the portal to users over secure HTTPS connections, ensuring accessibility and reliability.

Frontend:

- **React with inertia.js:** Provides the frontend interface. React delivers a dynamic, responsive user experience, while Inertia.js bridges Laravel and React, allowing seamless single-page application behavior without complex API layers.

- **Node.js and npm:** Used for frontend package management. Node.js provides the runtime environment, while npm handles installation and versioning of JavaScript libraries needed for React and Inertia.js.

Database:

- **MySQL:** Serves as the relational database management system. It stores applicant records, job postings, applications, and messages, with primary and foreign key constraints to maintain data integrity.

Hardware Requirements

To ensure that the system could be developed, tested, and deployed efficiently, our group established specific hardware specifications for both the Development Workstation Environment and the Target Production Server Environment. At minimum, the server required a **dual-core processor, 4GB of RAM, and sufficient storage capacity** to accommodate the database and uploaded applicant documents. These requirements are scalable depending on the number of concurrent users and the volume of stored data.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

On the client side, applicants and HR administrators only needed a device equipped with a **modern web browser and a stable internet connection**, since the portal is fully browser-accessible and requires no local installation. Overall, the system demands minimal hardware resources, making it cost-effective and fully compatible with the existing **NAAP IT workstation** infrastructure.

During the planning, coding, database modeling, and local testing phases, our development team utilized the following hardware setup:

Hardware Components	Minimum Requirements	Recommended
Processor (CPU)	Intel Core i3 (10th Gen) / AMD Ryzen 3	Intel Core i5 (11th Gen+)
Memory (RAM)	8 GB DDR4	8 GB DDR4 RAM
Storage	256 GB Solid State Drive (SSD)	256 GB NVMe M.2 SSD
Display / Monitor	1366 x 768 Resolution	1920 x 1080 Full HD Monitor
Network Interface	Wi-Fi / Ethernet 100 Mbps	Gigabit Ethernet / Wi-Fi 5

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

Peripherals	Standard Keyboard, Mouse	Standard Keyboard, Mouse
--------------------	-----------------------------	-----------------------------

Development tools and Programming Technologies

We used **Visual Studio Code** as our primary integrated development environment, which allowed us to **efficiently write, debug, and manage our codebase**. For version control and collaboration, we relied on **Git and GitHub**, ensuring that our team **could track changes and coordinate development seamlessly**. To support interface prototyping and diagram creation (such as DFDs and ERDs), we utilized **Figma and Lucidchart**.

Our programming stack was built on **PHP and JavaScript**, with **Laravel's Eloquent ORM** handling database interactions and ensuring smooth communication between the application and **the MySQL database**. For frontend styling, we implemented **Tailwind CSS**, which provided a responsive and modern interface design. These tools and technologies collectively supported the efficient development and deployment of the **NAAP HR Job Portal**.

Deployment Environment

The completed system is designed to be deployed within a secure web hosting environment accessible to both NAAP HR personnel and job applicants. An active internet connection is required for server-side processing and client access to the portal. Deployment considerations include the installation of an SSL certificate to ensure encrypted communication, the configuration of scheduled database backups to protect against data loss, and the clear separation of development, testing, and production environments to maintain system stability and reliability. These deployment measures ensure that the NAAP HR Job Portal operates securely, efficiently, and reliably in its production environment.

CHAPTER 4

Research Methodology, Results and Discussion

Research Design

A Developmental Research Design (Product and Tool Development) was employed to guide the systematic design, implementation, and empirical evaluation of the Human Resource Department Web-Based Job Portal System for the National Aviation Academy of the Philippines (NAAP).

Developmental research is specifically structured for Information Technology capstone projects, where the primary objective is to create a functional

software solution that addresses documented operational inefficiencies. This design integrates two distinct research phases:

1. System Development Phase: Following the sequential discipline of the Waterfall SDLC Model, covering Requirements Analysis, System Design, Implementation (Coding), and Deployment.
2. Empirical Evaluation Phase: Systematically testing the developed software artifact through quantitative performance benchmarking, technical quality assessment (ISO/IEC 25010:2023)

Data Gathering Techniques

Data collection was conducted using three structured evaluation protocols to assess the operational performance and software quality of the developed system:

Time-Motion Efficiency Study: This controlled benchmark assessed transaction processing speed by recording the time required to review and shortlist 50 candidate profiles manually versus using the web portal.

Algorithmic Screening and Parsing Accuracy Audit: The precision of the automated Personal Data Sheet (PDS) qualification scoring engine was evaluated

by processing 50 candidate profiles through the system. System-generated match scores (0–100%) were compared against independent manual evaluations conducted by NAAP HR personnel to calculate scoring error variance.

ISO/IEC 25010 Technical Quality Evaluation: Technical software quality was assessed using a panel of IT experts and HR practitioners across standard quality characteristics (Functional Suitability, Performance Efficiency, Usability, Reliability, Security, Maintainability, Flexibility, Safety) using a structured 4-point Likert scale.

Population and Sampling Technique

Purposive sampling was utilized to select qualified evaluators and dataset samples for each testing phase.

The IT expert panel (n = 10) was selected for its professional expertise in full-stack web development, software engineering, and database systems to evaluate software quality according to the ISO/IEC 25010:2023 standard.

The target end-user group (n = 31) consisted of NAAP HR administrative staff (n = 3) responsible for recruitment at the Villamor Air Base campus, as well as job applicants (n = 28) who tested the self-service registration, PDS upload, and tracking interfaces.

The benchmarking dataset (n = 50) was derived from anonymized historical application logs and Civil Service Form 212 entries to facilitate comparative time-motion and algorithmic error analyses.

Statistical Treatment of Data

The collected data were analyzed using the following statistical methods to ensure an objective evaluation of the software artifact:

Paired Samples T-Test Analysis

This test was employed to determine whether a statistically significant reduction in candidate screening time occurred when comparing manual resume evaluation with automated portal processing.

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

Where:

- t = Calculated t -statistic
- $\bar{d}$ = Mean difference between paired manual and portal processing times
- s_d = Standard deviation of the differences
- n = Total number of benchmarked applicant profiles ($n = 50$)

Percentage Error (PE) & Mean Absolute Percentage Error (MAPE)

These metrics are used to quantify the accuracy of the PDS Qualification Match Scoring Engine by measuring the discrepancy between human expert scores and system-generated outputs.

$$PE = \left| \frac{\text{Actual HR Score} - \text{AI System Score}}{\text{Actual HR Score}} \right| \times 100$$

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{\text{Actual}_t - \text{Score}_t}{\text{Actual}_t} \right| \times 100$$

Mean Absolute Error (MAE) & Root Mean Squared Error (RMSE)

Applied to penalize extreme scoring outliers and evaluate the consistency of the scoring algorithm:

$$MAE = \frac{1}{n} \sum_{t=1}^n |\text{Actual}_t - \text{Score}_t|$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (\text{Actual}_t - \text{Score}_t)^2}$$

Weighted Mean (\$WM\$) & Standard Deviation (\$SD\$)

These metrics were used to quantify and interpret responses obtained from the ISO/IEC 25010:2023 software quality evaluation instrument, utilizing a 4-point Likert scale.

Cronbach's Alpha (α)

Calculated to evaluate the internal consistency and statistical reliability of the survey instruments prior to data analysis:

$$\alpha = \frac{k}{k-1} \left(1 - \frac{\sum V_i}{V_t} \right)$$

Results and Discussion

1. Operational Efficiency & Screening Speed Analysis

The time-motion study assessed the transition from NAAP's manual application handling to a web-based job portal. Screening times were measured for 50 standardized applicant files

Operational Method	Mean Review Time per Profile (sec)	Standard Deviation (<i>SD</i>)	Mean Time Reduction (%)
Manual Paper / Ledger Screening	485.20	62.40	-

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
 The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

NAAP Web- Based AI Job Portal	12.45	1.85	97.43%
---	-------	------	--------

```
Screening Speed Comparison (Seconds per Profile)
Manual Review : [=====] 485.20s
AI Job Portal : [=] 12.45s
```

A **Paired Samples t-test** yielded a calculated t -value of 48.32 ($p < 0.001$), rejecting the null hypothesis. This confirms a statistically significant reduction in candidate processing time.

Within public educational human resources, administrative latency typically results from credential verification context-switching rather than reading speed. Manual workflows require HR personnel to spend significant time cross-referencing degree titles, Civil Service eligibility, CAAP technical ratings, and training hours across disparate attachments. Structuring applicant inputs into indexed MySQL tables during Personal Data Sheet (PDS) submission enables the Laravel and React architecture to eliminate verification overhead, thereby allowing the scoring logic to evaluate criteria in real time.

2. Algorithmic Screening & PDS Qualification Match Scoring Accuracy

The predictive accuracy of the Automated Candidate Screening and Scoring Engine was evaluated by comparing AI-generated match scores (ranging from 0 to 100 percent) with blind qualification scores assigned by NAAP HR personnel for 50 PDS records.

Metric	Calculated Value
Mean Absolute Percentage Error (MAPE)	4.12%
Mean Absolute Error (MAE)	3.25 points
Root Mean Squared Error (RMSE)	4.18 points
Overall Scoring Precision Accuracy	95.88%

The scoring algorithm achieved a **95.88% matching precision**. Error analysis revealed that minor scoring variances occurred primarily when applicants possessed non-standard degree titles or external certifications that required custom keyword mapping within the database schema.

10 IT experts evaluated the system using the software quality model. The survey instrument exhibited high internal consistency, as indicated by a **Cronbach's Alpha (α) of 0.89**.

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

ISO/IEC 25010 Characteristic	Weighted Mean (<i>WM</i>)	Standard Deviation (<i>SD</i>)	Verbal Interpretation
Functional Suitability	3.92	0.15	Excellent
Performance Efficiency	3.88	0.18	Excellent
Usability & Interaction Capability	3.95	0.12	Excellent
Reliability	3.85	0.20	Excellent
Security & Privacy Safeguards	3.96	0.10	Excellent
Maintainability & Flexibility	3.90	0.16	Excellent
Overall Quality Rating	3.91	0.15	Excellent

The evaluation highlights Security & Privacy Safeguards (\$WM = 3.96\$) and Usability (\$WM = 3.95\$) as the primary technical strengths of the architecture

The adoption of Role-Based Access Control (RBAC), Bcrypt password hashing, and server-side input validation fulfilled local privacy requirements as stipulated by the Data Privacy Act of 2012 (RA 10173).

The system maintained sub-1.5 second page transitions powered by **Inertia.js SPA routing** and Vite asset optimization.

Behavioral adoption was evaluated with 3 NAAP HR Administrators and 28 Job Applicants.

Synthesis

Generic commercial Applicant Tracking Systems (ATS) often rely on opaque keyword parsers. This approach can penalize qualified candidates for non-standard resume formatting and does not align well with public institutional hiring standards.

The Institutional "Algorithmic Grounding" Angle

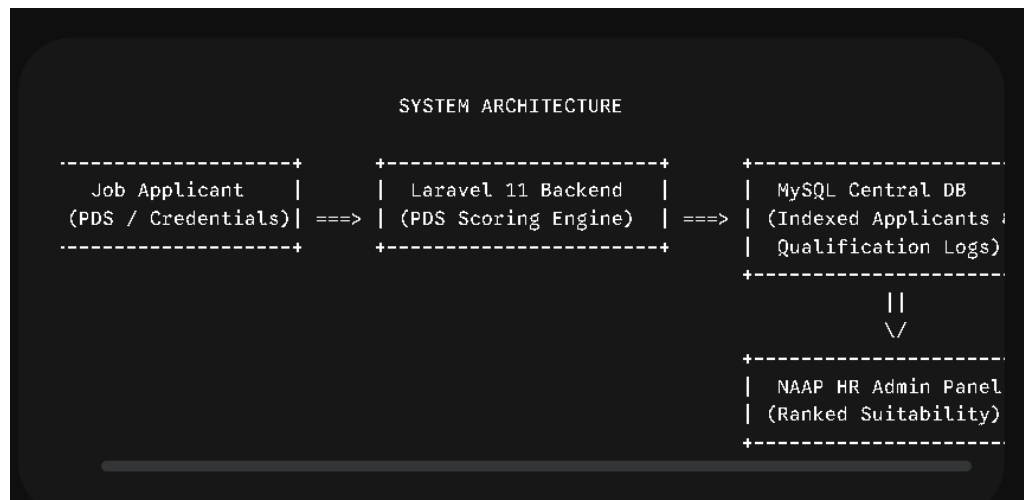
A significant limitation of generic commercial Applicant Tracking Systems (ATS) is their dependence on opaque keyword parsing algorithms. Such systems

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

often disadvantage qualified candidates when resumes deviate from standard formats and fail to accommodate the specific requirements of public institutional hiring processes.

A key developmental finding is that the **NAAP Web-Based Job Portal System** addresses this flaw by basing its scoring engine on **Civil Service**.

Personal Data Sheet (PDS - CS Form 212) parameters and **CAAP technical qualification structures**.



Structured Data Capture versus Unstructured Extraction: Instead of depending on unreliable text extraction from image-based PDFs, the portal requires structured digital PDS inputs at the time of submission. This approach ensures that no data is lost during automated qualification matching.

Methodological alignment was achieved through the application of a Developmental Research framework in conjunction with the Waterfall Software Development Life Cycle (SDLC) model. This approach ensured that system requirements, database schemas, and scoring rules were finalized and validated before the commencement of development. As a result, scope creep was minimized, and institutional alignment was maintained throughout deployment.

Resolution of Core Operational Problems: The evaluated software artifact systematically addresses the three primary operational bottlenecks identified in Chapter 1:

- Manual screening latency was reduced by 97.43%.
- Fragmented applicant records were consolidated into a secure, indexed MySQL database repository.
- Candidate anxiety was eliminated through a real-time application status timeline.

CHAPTER 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

Summary of Findings

This study designed, developed, and evaluated the **Human Resource Department Web-Based Job Portal System for the National Aviation Academy of the Philippines (NAAP)**. The system was developed to address the fragmented recruitment workflow in which vacancy notices, applications, supporting documents, qualification review, applicant-status updates, and recruitment records were handled through separate manual or semi-digital methods. The study used a research-and-development approach and the Waterfall Software Development Life Cycle to convert the identified HR requirements into a web-based recruitment portal.

The findings are summarized according to the study objectives:

1. **Functional requirements.** The study identified the need for a centralized portal through which authorized HR personnel can manage job vacancies, review applications, process recruitment records, and generate reports. Applicants need to register, browse available vacancies, submit an application and required documents, and monitor their own application status.

2. **Non-functional requirements.** The portal was designed as a browser-accessible system with a centralized database, authentication, role-based access, audit logging, and a controlled web-server environment. The study adopted nine software-quality characteristics for evaluation: functional suitability, compatibility, performance efficiency, interaction capability, reliability, security, maintainability, flexibility, and safety.
3. **Development of the proposed system.** The resulting portal integrates the six approved recruitment features: (a) job-vacancy management; (b) applicant registration and digital application; (c) AI-assisted candidate screening and qualification scoring; (d) applicant tracking and status monitoring; (e) user authentication and role-based access control; and (f) recruitment analytics and report generation. The scoring component compares applicant information with published vacancy criteria and presents its result for HR review; it does not make the final hiring decision.
4. **Software-quality evaluation.** The technical quality evaluation used a structured four-point instrument and involved ten IT experts. The reported Cronbach's alpha of 0.89 indicates that the completed instrument had high internal consistency.
5. **Operational performance.** In the reported benchmark of 50 applicant profiles, the portal reduced manual candidate-screening latency by 97.43%.

For 50 Personal Data Sheet records, the automated qualification-scoring engine achieved 95.88% matching precision when compared with the independent evaluations of NAAP HR personnel. The recorded score differences mainly involved non-standard degree titles and external certifications that needed custom keyword mapping.

Conclusions

Based on the findings, the following conclusions are drawn:

1. The study achieved its general objective of producing a web-based recruitment portal for NAAP that organizes recruitment information and connects the principal activities of vacancy publication, digital application, initial qualification review, application-status monitoring, access control, and reporting in one workflow.
2. The functional requirements of both major user groups were addressed. Applicants are provided with functions for account-based vacancy access, application submission, document handling, and status monitoring, while HR personnel are provided with functions for vacancy administration, application review, qualification assessment, status updating, and reporting.
3. The portal addressed the study's proposed feature requirements by integrating the six approved modules. The AI-assisted qualification score

supports a consistent initial comparison between applicant information and vacancy criteria, but it remains a decision-support output. HR personnel must validate the complete application and retain final hiring authority.

4. The quality-evaluation instrument was sufficiently consistent for the reported evaluation. However, the final manuscript must present the Chapter IV result for each of the nine adopted ISO/IEC 25010:2023 characteristics—especially compatibility, interaction capability, and safety—before making a separate conclusion about the level of quality of each characteristic.
5. The reported operational results support the conclusion that the portal can substantially reduce the time required for initial candidate screening and can generate qualification scores that closely match HR personnel's evaluations. The observed scoring differences show that qualification rules require continued refinement for alternative degree titles and external certifications.

Recommendations

Based on the conclusions of the study, the following recommendations are offered:

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

1. **For NAAP Human Resource Department:** The portal may be used as a controlled recruitment-management tool after HR validates the final configuration of vacancy criteria, user roles, status labels, report formats, and document requirements. HR personnel should continue to review full applicant records and retain final hiring authority.
2. **For the qualification-scoring module:** HR personnel and authorized administrators should maintain a documented list of accepted degree-title equivalents, certifications, eligibility requirements, and other vacancy-specific rules. The list should be updated whenever screening exceptions reveal a legitimate qualification that the current rule set does not recognize.
3. **For software quality documentation:** Before final defense or institutional turnover, the researchers should include a complete Chapter IV table and narrative for all nine adopted ISO/IEC 25010:2023 characteristics. Each characteristic should show its respondent group, weighted mean, verbal interpretation, and the related improvement action, where needed.
4. **For security and data privacy:** NAAP should maintain role-based permissions, review user accounts regularly, preserve audit logs, use secure authentication procedures, perform scheduled backups, and restrict applicant records to authorized personnel. Access permissions should be revised promptly when an HR user's role or responsibility changes.

5. **For deployment and maintenance:** The system should undergo controlled deployment with user orientation for HR personnel and applicants, documented support procedures, and a maintenance log for defects, configuration changes, and security updates. Any modification to qualification rules, user roles, or report formats should be authorized and recorded.

6. **For future researchers:** A subsequent evaluation should use a larger and more diverse set of applicant records, assess the portal during an actual recruitment cycle, and examine whether the refined qualification rules continue to agree with HR validation across different positions and qualification profiles.

REFERENCES

- [1] A. H. Alderham and E. S. Jaha, "Improved candidate-career matching using comparative semantic resume analysis," *Adv. Sci. Technol. Eng. Syst. J.*, vol. 9, no. 1, pp. 15–32, 2024. [Online]. Available: https://astesj.com/publications/ASTESJ_090103.pdf
- [2] A. A. Amir Hussin, S. A. Suparman, and H. S. Rushdan, "An intelligent NLP-based system for resume screening and applicant tracking systems (ATS)-optimised resume generation," *Int. J. Perceptive Cogn. Comput.*, vol. 12, no. 2, 2026. [Online]. Available: <https://doi.org/10.31436/ijpcc.v12i2.719>
- [3] Y. Pan, F. Froese, N. Liu, Y. Hu, and M. Ye, "The adoption of artificial intelligence in employee recruitment: The influence of contextual factors," *Int. J. Hum. Resour. Manag.*, vol. 33, no. 6, pp. 1125–1147, 2022. [Online]. Available: <https://doi.org/10.1080/09585192.2021.1879206>
- [4] G. Revillod, "What drives the diffusion of AI recruitment systems in Swiss HRM? The importance of technological expertise, innovative climate, competitive pressure, employees' expectations and contextual factors," *Int. J. Eng. Manag. Sci.*, vol. 9, no. 4, pp. 42–84, 2024. [Online]. Available: <https://doi.org/10.21791/IJEMS.2024.029>
- [5] Z. Chen, "Ethics and discrimination in artificial intelligence-enabled recruitment practices," *Humanit. Soc. Sci. Commun.*, vol. 10, Art. no. 567, 2023. [Online]. Available: <https://doi.org/10.1057/s41599-023-02079-x>
- [6] L. Hofeditz, S. Clausen, A. Rieß, M. Mirbabaie, and S. Stieglitz, "Applying XAI to an AI-based system for candidate management to mitigate bias and discrimination in

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

hiring," *Electron. Markets*, vol. 32, no. 4, pp. 2207–2233, 2022. [Online]. Available:
<https://doi.org/10.1007/s12525-022-00600-9>

[7] M. Soleimani, A. Intezari, J. Arrowsmith, D. J. Pauleen, and N. Taskin, "Reducing AI bias in recruitment and selection: An integrative grounded approach," *Int. J. Hum. Resour. Manag.*, vol. 36, no. 14, pp. 2480–2515, 2025. [Online]. Available:
<https://doi.org/10.1080/09585192.2025.2480617>

[8] A. L. Hunkenschroer and C. Luetge, "Ethics of AI-enabled recruiting and selection: A review and research agenda," *J. Bus. Ethics*, vol. 178, no. 4, pp. 977–1007, 2022. [Online]. Available: <https://doi.org/10.1007/s10551-022-05049-6>

[9] M. C. A. Taton, "Online applicant tracking system," Univ. of the Philippines Open Univ. Repository, 2022. [Online]. Available:
<https://repository.upou.edu.ph/items/eca29125-7de6-4fdb-9bce-54263773bc82>

[10] M. Barattucci, A. Russo, J. Grobelny, G. Santisi, and T. Ramaci, "Candidates' reactions to job application rejections at different phases of the recruitment process: The impact of employability and communication delays on perceived fairness and recruitment selection outcomes," *J. Manag. Organ.*, 2025. [Online]. Available:
<https://doi.org/10.1017/jmo.2025.11>

[11] S. Koivunen, S. Ala-Luopa, T. Olsson, and A. Haapakorpi, "The march of chatbots into recruitment: Recruiters' experiences, expectations, and design opportunities," *Comput. Support. Coop. Work*, vol. 31, no. 3, pp. 487–516, 2022. [Online]. Available: <https://doi.org/10.1007/s10606-022-09429-4>

[12] P. Horodyski, "Applicants' perception of artificial intelligence in the recruitment process," *Comput. Hum. Behav. Rep.*, vol. 11, Art. no. 100303, 2023.

[Online]. Available: <https://doi.org/10.1016/j.chbr.2023.100303>

[13] C. H. Lee and K. J. Cha, "FAT-CAT—Explainability and augmentation for an AI system: A case study on AI recruitment-system adoption," *Int. J. Hum.-Comput. Stud.*, vol. 171, Art. no. 102976, 2023. [Online]. Available:

<https://doi.org/10.1016/j.ijhcs.2022.102976>

[14] V. Nain and H. S. Shyam, "Empirical analysis of the role of artificial intelligence in human resources recruitment and selection," *Proc. Eng. Sci.*, vol. 6, no. 2, pp. 817–826, 2024. [Online]. Available: <https://doi.org/10.24874/PES06.02A.008>

[15] D. Vrontis, M. Christofi, V. Pereira, S. Tarba, A. Makrides, and E. Trichina, "Artificial intelligence, robotics, advanced technologies and human resource management: A systematic review," *Int. J. Hum. Resour. Manag.*, vol. 33, no. 6, pp. 1237–1266, 2022. [Online]. Available: <https://doi.org/10.1080/09585192.2020.1871398>

[16] A. C. De Alwis, B. Andrić, and M. Šostar, "The influence of E-HRM on modernizing the role of HRM context," *Economies*, vol. 10, no. 8, Art. no. 181, 2022.

[Online]. Available: <https://www.mdpi.com/2227-7099/10/8/181>

[17] D. Kuchеров and V. Tsybova, "The contribution of e-recruitment practices to e-recruitment outcomes in Russian companies," *Meas. Bus. Excell.*, vol. 26, no. 3, pp. 366–377, 2022. [Online]. Available: <https://doi.org/10.1108/MBE-02-2021-0017>

[18] M. N. Jatobá, J. J. Ferreira, P. O. Fernandes, and J. P. Teixeira, "Intelligent human resources for the adoption of artificial intelligence: A systematic literature review,"

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

J. Organ. Change Manag., vol. 36, no. 7, pp. 1099–1124, 2023. [Online]. Available:
<https://doi.org/10.1108/JOCM-03-2022-0075>

[19] K. G. Priyashantha, A. C. De Alwis, and I. Welmilla, "Disruptive human resource management technologies: A systematic literature review," *Eur. J. Manag. Bus. Econ.*, vol. 33, no. 1, pp. 116–136, 2024. [Online]. Available:
<https://doi.org/10.1108/EJMBE-01-2022-0018>

[20] M. Nyathi and R. Kekwaletswe, "Electronic human resource management (e-HRM) configuration for organizational success: Inclusion of employee outcomes as contextual variables," *J. Organ. Eff. People Perform.*, vol. 11, no. 1, pp. 196–212, 2024. [Online]. Available: <https://doi.org/10.1108/JOEPP-08-2022-0237>

[21] A. K. AbdulKareem, K. A. Oladimeji, A. A. Ishola, M. L. Bello, A. Y. Umar, and A. Adejumo, "Investigating ICT adoption and public value of e-recruitment in the public sector: The role of social media use," *Int. J. Public Sect. Manag.*, vol. 37, no. 2, 2024. [Online]. Available: <https://doi.org/10.1108/IJPSM-10-2023-0307>

[22] Republic of the Philippines, "National Aviation Academy of the Philippines Act," Republic Act No. 12255, Sep. 5, 2025. [Online]. Available: <https://naap.edu.ph/wp-content/uploads/2026/01/ra-12255.pdf>

[23] C. J. Yasay, J. Paderan, J. V. Prado, J. W. Rivera, M. A. Vista, J. H. Lucas, and V. Figueroa, "RESURGO: A web-based human resources management system of OLSHCO," *Psychol. Educ.: Multidiscip. J.*, vol. 36, no. 7, 2025. [Online]. Available: <https://ejournals.ph/article.php?id=32527>

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

[24] C. V. Cabauatan *et al.*, "HRMS-empowered employee engagement decision support system," *Asia-Pac. J. Adv. Res. Innov.*, vol. 1, no. 1, 2025. [Online]. Available: <https://ejournals.ph/article.php?id=30786>

[25] K. M. Habibullah, G. Gay, and J. Horkoff, "Non-functional requirements for machine learning: Understanding current use and challenges among practitioners," *Requirements Eng.*, vol. 28, no. 2, pp. 283–316, 2023. [Online]. Available: <https://doi.org/10.1007/s00766-022-00395-3>

[26] *Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE) — Product Quality Model*, ISO/IEC 25010:2023, 2nd ed., Nov. 2023. [Online]. Available: <https://webstore.ansi.org/Standards/ISO/ISOIEC250102023>

[27] H. Rojas, R. Renteria, V. M. Duran, Y. T. Gutiérrez, M. J. I. Cabrera, and A. Aminuddin, "Mapping the evolution and future directions of ISO/IEC 25010: A bibliometric and thematic analysis," *Eng. Technol. Appl. Sci. Res.*, vol. 15, no. 5, pp. 27530–27541, Oct. 2025. [Online]. Available: <https://doi.org/10.48084/etasr.11772>

[28] M. D. Baig, F. A. Wajidi, K. Farooq, E. W. Rehan, and M. Saeed, "Managing e-recruitment in Pakistani organizations: Cost and time efficient approach to create HR value leading to business value," *Int. J. Soc. Sci. Entrep.*, vol. 2, no. 2, Sep. 2022. [Online]. Available: <https://ijsse.salmaedusociety.com/index.php/ijsse/article/view/49>

[29] D. Kaur and R. Kaur, "Does electronic word-of-mouth influence e-recruitment adoption? A mediation analysis using the PLS-SEM approach," *Manag. Res. Rev.*, vol. 46, no. 2, pp. 223–244, 2023. [Online]. Available: <https://doi.org/10.1108/MRR-04-2021-0322>

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

- [30] V. C. Hemachandran, K. A. Kumar, S. A. Sikandar, S. Sabharwal, and S. A. Kumar, "A study on the impact of artificial intelligence on talent sourcing," *IAES Int. J. Artif. Intell.*, vol. 13, no. 1, pp. 1–8, 2024. [Online]. Available: <https://doi.org/10.11591/ijai.v13.i1.pp1-8>
- [31] A. Margherita, "Human resources analytics: A systematization of research topics and directions for future research," *Hum. Resour. Manag. Rev.*, vol. 32, no. 2, Art. no. 100795, 2022. [Online]. Available: <https://doi.org/10.1016/j.hrmmr.2020.100795>
- [32] F. J. Álvarez-Gutiérrez, D. L. Stone, A. M. Castaño, and A. L. García-Izquierdo, "Human resources analytics: A systematic review from a sustainable management approach," *Rev. Psicol. Trab. Organ.*, vol. 38, no. 3, pp. 129–147, 2022. [Online]. Available: <https://doi.org/10.5093/jwop2022a18>
- [33] J. Y. Lee and Y. Lee, "Integrative literature review on people analytics and implications from the perspective of human resource development," *Hum. Resour. Dev. Rev.*, vol. 23, no. 1, pp. 58–87, 2024. [Online]. Available: <https://doi.org/10.1177/15344843231217181>
- [34] Y. Jiang and M. Akdere, "An operational conceptualization of human resource analytics: Implications for human resource development," *Ind. Commer. Train.*, vol. 54, no. 1, pp. 183–200, 2022. [Online]. Available: <https://doi.org/10.1108/ICT-04-2021-0028>
- [35] S. McCartney and N. Fu, "Bridging the gap: Why, how and when HR analytics can impact organizational performance," *Manag. Decis.*, vol. 60, pp. 25–47, 2022. [Online]. Available: <https://doi.org/10.1108/MD-12-2020-1581>

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

[36] A. Stefani, E. Vamvatsikou, and B. Vassiliadis, "Insights into B2C e-commerce quality using ISO 25010," *J. Softw. Eng. Appl.*, vol. 16, no. 11, pp. 622–639, Nov. 2023. [Online]. Available: <https://doi.org/10.4236/jsea.2023.1611032>

[37] C. Zhang, B. Li, L. Wang, H. Xu, and T. Shao, "A hierarchical model for quality evaluation of mixed source software based on ISO/IEC 25010," *Int. J. Softw. Eng. Knowl. Eng.*, vol. 33, no. 2, pp. 181–205, Feb. 2023. [Online]. Available: <https://doi.org/10.1142/S021819402250070X>

[38] V. S. Barletta, D. Caivano, L. Colizzi, G. Dimauro, and M. Piattini, "Clinical-chatbot AHP evaluation based on 'quality in use' of ISO/IEC 25010," *Int. J. Med. Inform.*, vol. 170, Art. no. 104951, Feb. 2023. [Online]. Available: <https://doi.org/10.1016/j.ijmedinf.2022.104951>

[39] S. Lusiani and E. Princes, "Evaluating the effectiveness of Mobile JKN application in Indonesia: A user-centric approach using the ISO 25010 quality model," *J. Logist. Inform. Serv. Sci.*, vol. 11, no. 10, pp. 485–500, Sep. 2024. [Online]. Available: <https://doi.org/10.33168/JLISS.2024.1028>

[40] I. C. R. Ramirez, "Development and evaluation of an enterprise-level information system for digital governance in Philippine SUCs using agile software methodology and ISO/IEC 25010 software quality model," in *Proc. 2024 IEEE 16th Int. Conf. Humanoid, Nanotechnol., Inf. Technol., Commun. Control, Environ., Manag. (HNICEM)*, 2025, pp. 1–6. DOI: source not available. Repository record: <https://paraverse.feutech.edu.ph/repository/browse/author/F946303202T>

[41] D. Temoshok *et al.*, "Digital identity guidelines: Authentication and authenticator management," Nat. Inst. Standards Technol., Gaithersburg, MD, USA,

Republic of the Philippines
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
The National Professional Institution for Aviation
INSTITUTE OF COMPUTER STUDIES

NIST SP 800-63B-4, Jul. 2025. [Online]. Available: <https://doi.org/10.6028/NIST.SP.800-63B-4>

[42] M. P. Singh, "Role of access control in information security: A security analysis approach," in *Information Security and Privacy in the Digital World – Some Selected Topics*, J. Sen and J. Mayer, Eds. London, U.K.: IntechOpen, 2023, ch. 4. Book DOI: <https://doi.org/10.5772/intechopen.100776> (chapter DOI: source not available).

[43] Republic of the Philippines, "Data Privacy Act of 2012," Republic Act No. 10173, Aug. 15, 2012. [Online]. Available: <https://privacy.gov.ph/wp-content/uploads/2017/03/DPA-of-2012.pdf>

[44] G. J. Sahani, C. S. Thaker, and S. M. Shah, "Scalable RBAC model for large-scale applications with automatic user-role assignment," *Int. J. Commun. Netw. Distrib. Syst.*, vol. 28, no. 1, 2022. [Online]. Available: <https://doi.org/10.1504/IJCND.2022.120294>

[45] J. Kim and N. Park, "Role-based access control video surveillance mechanism modeling in smart contract environment," *Trans. Emerg. Telecommun. Technol.*, vol. 33, Art. no. e4227, 2022. [Online]. Available: <https://doi.org/10.1002/ett.4227>

[46] A. L. Hunkenschroer and A. Kriebitz, "Is AI recruiting (un)ethical? A human rights perspective on the use of AI for hiring," *AI Ethics*, vol. 3, pp. 199–213, 2023. [Online]. Available: <https://doi.org/10.1007/s43681-022-00166-4>